

MOTHERS AND SONS: PREFERENCE FORMATION AND FEMALE LABOR FORCE DYNAMICS*

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This paper argues that the growing presence of a new type of man—one brought up in a family in which the mother worked—has been a significant factor in the increase in female labor force participation over time. We present cross-sectional evidence showing that the wives of men whose mothers worked are themselves significantly more likely to work. We use variation in the importance of World War II as a shock to women's labor force participation—as proxied by variation in the male draft rate across U. S. states—to provide evidence in support of the intergenerational consequences of our propagation mechanism.

Women's role in the United States economy has dramatically changed during the last century: whereas at the beginning of the century women tended to have low labor force participation rates and to exit the formal labor market upon marriage, today almost 50 percent of the labor force is female, more women than men complete college, and women increasingly combine family and career.

What factors are responsible for this profound transformation in the role that women play in the family and in the workplace? The explanations proposed depend on the time period, and they range from the liberating effects of new consumer durables that, as suggested by Greenwood, Seshadri, and Yorukoglu [2004], greatly decreased the amount of work required to run a household (e.g., washing machine, vacuum cleaner, etc.), to the revolutionary effect of the oral contraceptive (the pill) that, as argued by Goldin and Katz [2002], facilitated a woman's investment in her career by almost eliminating the chance of an accidental pregnancy. The expansion of the service sector with its attendant white-collar jobs or skilled-biased technological change

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is also thought to have greatly facilitated this transformation (see Goldin [1990] for this argument and Galor and Weil [1996] for a model that relates skill-biased technological change to fertility and labor choices).

In this paper we suggest a new and complementary channel. We argue that a significant determinant of the gradual but steady increase in women's involvement in the formal labor market was the increasing number of men who, over time, grew up with a different family model—one in which their mother worked. Growing up with a working mother, we believe, either influenced a man's preferences for a working wife or directly made him a better partner (say, by increasing his ability to cooperate and be productive in household work) for a working woman. The presence of this different type of man, in turn, made investing in market skills and becoming a working woman more attractive. As the number of working mothers increased, so did the proportion of men raised with this different family model, which then helped to increase the relative supply of working women of the following generation. In this way, women who worked set an example for their sons, and thus made it easier for the next generation of women to follow in their footsteps. Thus, the gradual transformation of the family—long considered a source of transmission of moral, religious, and cultural beliefs—itself acted as a propagation mechanism of change in women's role.¹

Why should men with working mothers differ from other men? One possibility is that they are less averse to having a working wife than other men. For example, their idea of sex roles and what the division of work should be may differ. More simply put, they may have different preferences. This would make it more likely that, *ceteris paribus*, a man whose mother worked marries a woman who will work than a man whose mother did not work. Alternatively, men may have similar preferences, and women may have similar propensities to work in the market, but men brought up by working mothers may have greater household productivity arising perhaps from a different attitude toward participating in housework. This would allow their wives to spend more time in market relative to home production.

We model the two stories sketched above in a simple dynamic

1. Of course, as more women joined the labor force, attitudes toward these women changed in society at large. Our argument does not preclude this additional transmission mechanism. We emphasize, both theoretically and empirically, however, the role played directly by having a working mother.

framework and show that both specifications give rise to similar dynamic consequences: a larger proportion of men with working mothers in a given generation leads to an increase in women's incentives to invest in market skills and to a greater proportion of women who work in the next generation. Our model is capable of generating both a steady state that is interior and one in which women do not work. Thus, once women are participating in the labor force while they raise children, any particular shock that increases these women's labor force participation will also have dynamic repercussions over several generations.

Our empirical work examines both the cross-sectional and dynamic implications of our hypothesis. Using several data sets, we show that the probability that a man's wife works is positively and significantly correlated with whether his mother worked, even after controlling for many other background characteristics of husband and wife that may have influenced his spouse's working behavior and that may be driving the positive correlation. Depending on the definition of working mother used (our definition necessarily varies with the data set), we find that having a working mother significantly increases the probability that a man's wife works; the magnitude of the effect ranges from 24 to 32 percentage points depending on the data set used.

We investigate the dynamic implications of our theory by exploring the intergenerational consequences of two different sources of variation in the proportion of men brought up by working mothers. We first make use of idiosyncratic differences across U. S. states in the impact of World War II on married women's labor supply to provide exogenous variation in the proportion of men raised by working mothers. We analyze the effect of World War II on the labor supply of the 1930–1935 cohort of women, a cohort that was too young to be directly affected by the war, but that was the right age to be affected by the change in the available pool of men a few decades later. We contrast the indirect effect of the war on this cohort with its direct effect on older cohorts and show that although the direct effect of World War II faded for the older cohorts over time, its indirect effect on the younger cohort persisted. By the time the cohorts reach the age of 45–50, there is no longer a direct effect of the war on the older cohorts' labor supply whereas the indirect effect on the 1930–1935 cohort is still present and strong: we find that a 10 percent higher mobilization rate is associated with 3.3 additional weeks worked by women 45–50 years old in 1980, which represents

almost 16 percent of the total increase in weeks worked on average by this age group between 1940 and 1980.

Our second source of variation comes from differences across U. S. states in the average fertility of working relative to nonworking women. Our theory implies that, for a given level of female labor supply, states with higher ratios in the average fertility of working relative to nonworking women should have greater female labor supply the following generation. We examine the relationship between this measure of relative fertility and the next generation's female labor supply across U. S. states over several decades and show that the correlation in the data is positive and significant as predicted by the theory.

In addition to the aforementioned papers that attempt to explain why women's labor supply has changed, there is a large empirical and historical literature on women's labor force participation (see Killingsworth and Heckman [1986] for a survey and Goldin [1990] for an extensive analysis of the change in women's role in the labor market since the beginning of the twentieth century). Smith and Ward [1985] explicitly measure the contribution of demand factors to the increase in female participation between 1950 and 1980. They find that about 60 percent of the increase in women's labor force participation may be attributed to the increase in real wages that took place over this time period with changes in other factors, such as the increasing level of schooling, and changes in gender role attitudes, accounting for the rest. Pencavel [1998] examines the more recent history from 1975 to 1994 and concludes that increases in own wages account for between one-quarter to a half of the increase in women's labor supply (depending on the generation), with the increased attractiveness of the marketplace relative to the household accounting for the rest.² More closely related to our work are several papers by Goldin. Goldin [1997], in particular, provides an illuminating account of how work, marriage, and family options have changed over time by studying four generations of women. Goldin [1991] also studies the consequences of World War II on women's labor force participation.

2. Olivetti [2001] studies the relative effect of the increase in the returns to labor market experience and of the decline in the gender wage gap on the change in married women's life cycle labor supply between 1970 and 1990. Jones, Manuelli, and McGrattan [2003] study the effects of the decline of the gender wage gap and of technological improvements in the production of nonmarket goods on female labor supply over the past three decades.

Our paper is also related to a recent literature that examines how preferences evolve within the family. Bisin and Verdier [2000] examine how the marriage market and the endogenous transmission of cultural preferences interact. Galor and Moav [2002] argue that the (genetic) transmission of preferences that favored quality over quantity of children helps understand the history of economic growth.

Our paper also belongs to a growing literature that is interested in the effects of how (and why) individuals sort in particular ways in marriage, interfamily interactions, and the consequences of this for the macroeconomy. Fogli [2000] studies the effect of different family arrangements on labor market outcomes. Kremer [1997] and Fernández and Rogerson [2001] examine the consequences of marital sorting for inequality, and Fernández, Guner, and Knowles [2001] examine the relationship between sorting, inequality, and growth.

There is also a literature in psychology and sociology that examines parental influence on children's attitudes toward spouses and the division of labor in the household. Since the work of Freud [1950], psychoanalytic theory of mate selection has claimed that individuals tend to choose spouses who are similar to their opposite-sex parents. Several studies, using a variety of methodologies, have tested the empirical validity of Freud's Oedipal theory and on the whole seem to find evidence that supports it in a variety of dimensions.³ Parental roles in the household also appear to affect their children's attitudes and behavior. Thornton, Alwin, and Camburn [1983], for example, find that parental gender-role attitudes, education, and experiences affect their children's gender-role attitudes. In particular, a mother's attitude toward women working in the market relative to the home (itself correlated with her work experience after marriage) is associated with her children's attitudes toward the same at the age of eighteen. In the economics literature, Del Boca, Locatelli, and Pasqua [2000] use data from the 1995 Bank of Italy Survey to study whether husbands' employment status affects wives' labor force participation decisions.⁴ They find both the mother and mother-

3. See Daly and Wilson [1990] and Epstein and Guttman [1984] for reviews of the literature.

4. We find that only the mother-in-law is a significant determinant. We are able to control, however, for a large number of factors that the authors leave out and furthermore focus on whether the mother worked while her son was growing up.

in-law to be significant determinants of whether a son's wife works. Neumark and Postlewaite [1998] study the importance of relative income concerns on women's employment decisions. They find that women's employment decisions are significantly affected by those of their sisters and sisters-in-law.

This paper is organized as follows. In the next section we develop a simple dynamic model that formalizes our main idea. In the third section we examine the cross-sectional evidence, and in the fourth section we present the dynamic empirical evidence in favor of our hypothesis. The last section concludes.

I. THE MODEL

The objective of this section is to develop a simple dynamic model that captures the main elements of our general idea. To begin with, our model needs to provide a reason for why married women are more likely to work, *ceteris paribus*, if their husband's mother worked. The answer must be that either these men marry women who are, *ex ante*, more likely to work, or that there is something different about these households *ex post*, such that a woman in this type of household is more likely to end up working. Both explanations are likely to play a role, and both rely on these men differing, somehow, from their counterparts whose mothers did not work. The difference, in general, must lie either in "preferences" or in "technology/endowments." That is, either a son's tastes or attitudes are affected by having a working mother in such a way that his wife is (either *ex ante* or *ex post*) more likely to work, or the son ends up with a different set of household skills (or attitudes toward housework) that makes him a better partner (again, *ex ante* or *ex post*) for women who work.⁵ We show that both channels generate similar dynamics.

In our model of the preference channel, all men have the same endowments, but they differ in their preference for a working wife. If men with working mothers, *ceteris paribus*, are more likely to marry women who have high market productivity and hence are more likely to work, the greater presence of these men

5. These skills are most likely themselves a function of having different preferences. That is, it is doubtful that using a vacuum cleaner or washing machine requires a set of specialized skills, but rather that some men are more averse to engaging in these tasks. Furthermore, a preliminary look at data from the Panel Study of Income Dynamics (PSID) suggests that the hours a man spends on housework is independent of his mother's working behavior.

in the population will encourage women to invest more in market skills and lead to an increase in women's labor supply. Thus, in our preference channel we emphasize changes in marriage probabilities as the key ingredient leading to greater participation in the labor market.

In our model of the technology/endowment channel, men no longer differ in their attitudes toward their spouse's working behavior, but they differ in their household productivity or attitudes toward housework. If men with working mothers are more productive in the household, their spouses will spend more time working in the market than would otherwise identical women married to men with low household productivity. For constant marriage probabilities, the greater presence of these men in the population will encourage women to increase their investment in market skills (as they will have greater opportunity to use these), leading to greater female labor supply. Thus, in our household productivity channel we emphasize *ex post* differences among households as the key ingredient leading to greater female labor participation.

Below we illustrate how these channels might work without devoting great space to them as our empirical work will not attempt to investigate the reasons why these men differ, but rather to establish that they do and to show that these differences have quantitatively important dynamic repercussions.

I.A. The General Framework

We start with a common basic framework for both explanations and then make simplifications that allow us to highlight the workings of each channel with a minimum of algebra. The timing in the model is as follows. We assume that individuals live for two periods. In the first period, given the distribution of male types, each woman decides how much of an effort to make in investing in market skills. In the second period, men and women match, decide whether to marry or stay single, have children (if married), and make time-allocation decisions. Below we describe each stage of the decision-making process, starting with the time-allocation problem within a household.

I.B. Household Time Allocation

Assume that all individuals are endowed with a unit of time. Within a married household, each spouse decides how much of that time t to allocate to market activity; the remainder of the

time $(1 - t)$ is allocated to household production (for simplicity, we abstract from leisure). We assume that these time allocations are a Nash equilibrium of a game in which each spouse decides her or his time allocation taking as given the time allocation of the other partner.

Suppose that the utility function of a married man is given by

$$(1) \quad V_m(c, b, t_f; q_m) = c + \beta \log b - \alpha t_f + q_m,$$

where $\beta > 0$, c is the household consumption of the market good, b is the household good (services, quality of children, etc.), and q_m is the quality of the match between the man and his wife as perceived by the man. We will assume that men may directly dislike having their wives work.⁶ This is a modeling shortcut for an outcome that would arise endogenously if not all consumption (more generally, not all utility) in the household were known to be joint forever and this led a married woman to work in the market for more hours than would maximize the utility of her husband.⁷ Thus, $\alpha \geq 0$ is a parameter that we will later allow to vary across men in a systematic fashion.

A married woman's utility function is the same as a married man's except that there is no direct disutility from his market time and the quality of the match is now given by q_f ; i.e.,

$$(2) \quad V_f(c, b; q_f) = c + \beta \log b + q_f.$$

We assume that household consumption is joint and equal to the sum of each spouse's market earnings:

6. A Gallup poll conducted in 1938 asked, "Do you approve or disapprove of a married woman earning money in business or industry if she has a husband capable of supporting her?" A resounding 81 percent of men responded negatively [Erskine 1971]. The same question posed by the General Social Survey (GSS) showed that this fraction had declined to 38 percent of the white male population by 1972, 25 percent in 1982, and 17 percent in 1998.

An alternative possibility would be to assume that men like women who are similar to their mothers, e.g., if their mother worked, then they also like women who work; if their mother did not work, then they like women who do not work. This can be modeled as having men draw from different Q distributions depending on their mother's work behavior.

7. There are many reasons why all consumption may not be joint. Spouses, for example, may directly have different preferences over consumption bundles. If how income is allocated over different consumption goods is the outcome of intrafamily bargaining, this may lead women to work more than is otherwise optimal in order to ensure that their bargaining power does not erode too much over time (see, e.g., Lundberg and Pollak [2001]). Alternatively, all utility while married may be joint, but women may work more than is optimal from a man's perspective, to ensure that, in the case of divorce, her earnings potential is not too low as a result of little work experience. See our NBER 2002 working paper for a model that endogenizes a husband's disutility from a working wife.

$$c = w_m t_m + w_f t_f,$$

where w_m denotes the market wage (equivalently, the market productivity) of the husband and w_f that of the wife. The household good is produced with the following technology:

$$b = a_m h(1 - t_m) + a_f h(1 - t_f),$$

where the $a_i \geq 0$ are productivity parameters and h is an increasing strictly concave function with $h(0) = 0$.

To focus the discussion, suppose that the man has a comparative advantage in market work; i.e.,

$$\frac{w_m}{a_m} \geq \frac{w_f}{a_f}.$$

It is easy to see that this implies that the wife will put more time into household production than her spouse. We will assume throughout that parameters are such that at least one spouse works in the market (i.e., $w_m/a_m > \beta(h'(1)/a_m h(1) + a_f h(1))$). The first-order conditions then yield

$$(3) \quad \begin{aligned} \frac{w_m}{a_m} &\geq \beta \frac{h'(1 - t_m)}{a_m h(1 - t_m) + a_f h(1 - t_f)} \\ \frac{w_f}{a_f} &\leq \beta \frac{h'(1 - t_f)}{a_m h(1 - t_m) + a_f h(1 - t_f)}, \end{aligned}$$

since $h(0) = 0$ guarantees that at least some time is spent on home production. Hence, the solution either has both spouses devoting time to home and market production or, if a spouse is at a corner, then a woman who is specialized only works at home whereas a man who is specialized only works in the market. To further simplify the exposition, we will assume henceforth that all men share the same market productivity w_m .

It is easy to show that, *ceteris paribus*, women supply more hours the greater their market wage and the higher is their mate's household productivity. Henceforth we denote by V_{ij} the utility of an agent i married to agent j in which the time allocation of each spouse is the solution to the married individual's optimization problem.

If agents remain single, they also allocate their unit of time between market production and single home production. The utility from being single is assumed to be given by

$$(4) \quad u(c, b) = \max_t c + \gamma \log b$$

subject to

$$c = w_i t, \quad b = a_i h(1 - t), \quad 1 \geq t \geq 0,$$

where $\gamma > 0$. Henceforth we denote by $U(w_i, a_i)$ the indirect utility associated with the solution to the optimization problem (4).

I.C. Matching

We use the most basic matching model and assume that there is only one round of matching. Agents meet each other at random, and each obtains an iid match quality draw $q_i \in [\underline{q}, \bar{q}]$ from a continuous distribution Q . Agents either marry (if both of them prefer to marry than to remain single), or remain single.

For every market productivity level of the female, there is a cutoff quality, $q_m^*(w_m, w_f; \alpha, a_m, a_f)$, of the man for the woman, and another one, $q_f^*(w_m, w_f; \alpha, a_m, a_f)$, of the woman for the man, such that only matches above both cutoff qualities become marriages. This cutoff quality is simply the level of q that makes agent i indifferent between marrying individual j and remaining single; i.e., that equates $U(w_i, a_i)$ to $V_{ij}(w_i, w_j, \alpha, a_i, a_j, q_i)$.

I.D. Market Skill Acquisition

In the first stage of life, ex ante identical women decide how much effort e to make in obtaining market skills. This effort determines the distribution $W(w_f; e)$ from which they then draw a level of market productivity $w_f \in [w_f, \bar{w}_f]$. A greater effort generates a better distribution (in the sense of first-order stochastic dominance); i.e., we assume that $W(w_f; e_2) \leq W(w_f; e_1) \forall w_f$ if $e_2 > e_1$ and that W is a continuous function of e . We assume that the level of market skill is observable. These market skills can be thought of as the type (and not only quantity) of education a woman acquires. For example, it is argued that as more women entered the workplace they studied a different set of subjects and materials than previously.⁸

There is a convex, continuous, and increasing disutility from

8. Our cross-sectional results show that a man's wife is more likely to work if his mother worked, even after controlling for the wife's education. This is still compatible with women being ex ante different as long as years of education is not the only component of market skills, which is probably the case.

acquiring skills, $C(e)$, with $C(0) = 0$ and $C'(0) = 0$. A young woman will choose e to maximize her expected utility taking into account her marriage probabilities, the utility from marriage and the utility from remaining single.

The above constitutes our general framework. We now proceed by making a series of simplifying assumptions that allow us to first highlight the workings of the preference channel and then the workings of the household productivity channel.

II.E. The Preference Channel

Assume that all individuals have the same household productivity $a_f = a_m = 1$ and that h is linear, so that $b = 2 - (t_m + t_f)$. Furthermore, let $w_m > \beta$, and assume that men are always more highly paid in the workplace than women, i.e., $w_m > w_f$, $\forall w_f$ (if they were not, then in those households in which the inequality were reversed, the woman would work more hours than her spouse, but this would not alter our analysis otherwise). In this case, (3) implies that the husband works exclusively in the market. His wife, on the other hand, works exclusively at home if $w_f \leq \beta$, and works $t_f = 1 - \beta/w_f$ in the marketplace and the remainder of time at home if $w_f > \beta$.

To further simplify matters, we assume that the female wage distribution consists of only two outcomes, w_h and w_l , with $w_h > \beta > w_l$. With this wage profile, women who draw the high wage work both at home and in the workplace, whereas women who draw the low wage work exclusively at home.⁹ Greater effort increases, at a decreasing rate, the probability $\pi(e)$ with which the higher wage w_h is drawn, i.e., $\pi' > 0$, $\pi'' < 0$, and we assume that $\pi'(0) = \infty$ and $\pi'(\infty) = 0$.

A man's disutility from a working wife is assumed to depend on whether his mother worked outside the home. As a normalization, we let $\alpha = 0$ if a man's mother worked outside the home; otherwise α is some strictly positive level. This simply can be a reflection of Freudian behavior (men like women who resemble their own mothers), or it may arise because it challenges the conventional idea of sex roles in a household (e.g., identity as in Akerlof and Kranton [2000]). Thus, when we refer to a man's "type," what we mean is whether his mother worked in the

9. The analysis is not altered if $w_l > \beta$. The assumption that it is strictly smaller simplifies the exposition, however, as it implies that both types of men have the same cutoff quality for a woman who draws w_l .

market or not. We henceforth index a man's type by $i = h, l$ but where the i now refers to whether his mother had productivity w_h and therefore worked in the market or productivity w_l and worked exclusively at home.

Let $V^j(\lambda)$, $j = h, l$ be a woman's expected utility conditional upon drawing a wage w_j and given that a fraction λ of men of the same generation had working mothers. $V^j(\lambda)$ can be expressed as

$$(5) \quad \begin{aligned} V^h(\lambda) &= \lambda[p_{hh}V_h + (1 - p_{hh})U_h] + (1 - \lambda)[p_{hl}V_h + (1 - p_{hl})U_h] \\ V^l(\lambda) &= \lambda[p_{lh}V_l + (1 - p_{lh})U_l] + (1 - \lambda)[p_{ll}V_l + (1 - p_{ll})U_l], \end{aligned}$$

where V_j is the utility of being married, and U_j is the utility of being single, for a woman with market wage w_j , $j = h, l$. We have assumed that the probability of matching with a given type of men is equal to his proportion in the population. Note that in the female's utility from being married we have suppressed the male subscript. This is because, in our simple model, all men have the same market wage and household productivity and their type does not influence their choice of t_m . Consequently, a woman's own time allocation decision and hence also utility from being married to a particular man is independent of the man's type. p_{jk} , $j, k = h, l$ gives the probability that a woman with market wage w_j matched with a man of type k (i.e., one whose mother's productivity was w_k) married him, i.e.,

$$(6) \quad p_{jk} = \int_{q_j^*}^{\bar{q}} \int_{q_{kj}^*}^{\bar{q}} dQ dQ,$$

where q_j^* is a woman's reservation match quality given that she earns w_j and q_{kj}^* is a man's reservation match quality for a woman with productivity w_j given that he is of type k .

Note again that the reservation match quality of a woman only depends on her own type and not on the man's type since all men yield women the same utility (i.e., all men make the same time allocation decisions and have the same market productivity). Furthermore, note that since women with w_l do not work, $p_{lh} = p_{ll}$; i.e., men's reservation quality for these women does not depend on their own type. Lastly, note that $p_{hl} < p_{hh}$ since men whose mothers did not work obtain disutility from having a working wife.

Turning to a woman's investment problem, a young woman will choose e to maximize her expected lifetime utility:

$$(7) \quad \pi(e)V^h(\lambda) + (1 - \pi(e))V^l(\lambda) - C(e).$$

The first-order condition for the maximization problem (7) yields

$$(8) \quad \pi'(e)(V^h(\lambda) - V^l(\lambda)) - C'(e) \leq 0.$$

I.F. Comparative Statics

It is easy to show that at an interior solution for (8), the optimal e is increasing in λ ; i.e., $V^{h'} > V^{l'}$.¹⁰ These men make it more attractive to be a high market productivity woman since they are less likely to reject this type of woman if she wants to marry them. Consequently, a larger proportion of men with working mothers in the population implies a greater proportion of women with w_h relative to w_l , and hence more women working than before.

This is a convenient place to note that our results do not depend on the specific household utility assumed nor on the noncooperative solution we adopted. An alternative preference specification that permitted a husband to bargain with his wife on her labor supply would also yield the desired relationship between the amount a woman works and the working behavior of the husband's mother. As in the simpler model, an increase in the proportion of men with working mothers would make it more attractive for women to invest in market skills, since a high market skill woman would face a greater probability of being accepted in a match and of making use of her market skills once married.

I.G. The Household Productivity Channel

In the model described in the previous section, an increase in the proportion of men with working mothers increased the investment level in market skills by increasing the relative proportion of men who had low cutoff quality levels for working women. A different possibility is that men with working mothers are simply more productive in the household than men brought up by non-working mothers.¹¹ Marriage to one of these men would allow a woman to spend more time in the market than if married to a low

10. Note that an interior solution will exist since for $\lambda = 1$, e is interior given our assumptions on π and C and the solution to the maximization problem is continuous in λ .

11. Although we model the differences across men as productivity differences, we stress again that this could arise from different attitudes toward housework.

Cunningham [2001] uses a 31-year panel study of white mothers and children in the metropolitan Detroit, MI, area to examine parental influence on household work. He shows that the parental division of labor when a son was growing up affects the adult son's participation in routine housework once he marries.

household productivity man. This channel would give rise to similar comparative statics as the model above. We give a quick sketch of the mechanism below. To highlight the functioning of this channel, we now assume that men all have the same preferences and set $\alpha = 0$. Instead, men differ in a systematic fashion with respect to their household productivity.

As before, let men all share the same market productivity w_m . Female market productivity w_f is a random draw from a distribution that depends on the effort level exerted, as discussed in the general framework. We maintain our assumption that men have a comparative advantage in market work by assuming that $(w_m/\bar{a}_m) \geq \bar{w}_f$ (again, our analysis would go through if we relaxed this assumption). The household production is no longer assumed to be linear. We now assume instead that $h'(0) = \infty$, so that (3) implies that both partners work some time in the home.

As in the previous model, suppose that all women have the same household productivity $a_f = 1$, but that men's household productivity $a_m \in [\underline{a}_m, \bar{a}_m]$, is a random draw from a distribution that depends on whether his mother worked or not.¹² The idea here is that a man whose mother worked learned to cooperate and work in the household more than a man whose mother was exclusively a housewife. Slightly abusing notation, we now allow $j = h, l$ to index whether a man's mother's worked (h) or not (l) and assume that in the first case a man's household productivity is a random draw from $A_h(a_m)$ whereas, in the second case, it is a draw from $A_l(a_m)$. We assume that A_h first-order stochastically dominates A_l . Lastly, we abstract from the workings of the marriage market which was key in the previous section on preferences by assuming that match quality is sufficiently high that all matches are accepted.

Note that both first-order conditions in (3) are now met as strict equalities. Furthermore, a wife's labor supply to the market is increasing in her partner's household productivity. It follows that an increase in λ will increase women's investment in market skills since they will, on average, spend more time working in the market the greater is the proportion of men with working mothers. An increase in λ implies that there are more men with relatively high household productivity, which allows married

12. More generally, a man's household productivity draw could depend on how much his mother worked rather than on the zero-one outcome of not work/work, but this assumption simplifies notation.

women to put more hours into working in the market (i.e., these men function as women's "engine of liberation"). Since women know that their market skills will play a larger role, they will now invest more in these, compounding the prior effect by increasing the proportion of women with high market productivity. Thus, female labor supply increases both because a greater number of households have more men who are productive in the house and because women on average have higher market productivity as a result of their increased investment in market skills.

I.H. Dynamics

Both models discussed above give rise to the same qualitative dynamic evolution. Slightly abusing notation again, let $F_i(\lambda_t)$ be the number of married women of type $i = h, l$ given that the proportion of men with working mothers at time t is λ_t , where h now indicates that a woman worked outside the home and l that she did not.

Assuming that only married women have children, and that all women have the same number of children, the dynamics of the system are given by

$$(9) \quad \lambda_{t+1}(\lambda_t) = \frac{F_h(\lambda_t)}{F_h(\lambda_t) + F_l(\lambda_t)};$$

i.e., the proportion of men born to working women next period is simply the proportion of married working women in the married population in this period.

It is easy to see that λ_{t+1} is an increasing function of λ_t , iff,

$$(10) \quad \frac{1}{F_h} \frac{\partial F_h}{\partial \lambda} > \frac{1}{F_l} \frac{\partial F_l}{\partial \lambda};$$

i.e., iff an increase in λ produces a greater percentage increase in the number of married working women than in the number of married nonworking women.¹³

Condition (10) holds in both models sketched above. In the preference model, an increase in λ increases the number of women who work and marry both by increasing investment in market skills and by increasing the number of high market productivity women who marry. The number of married nonworking women falls, on the other hand, since although the marriage

13. Note that (10) is the condition that needs to hold even if there are fertility differentials across types of households.

probability of a low market productivity type remains constant, there are fewer of this type in the population.

In the household productivity model, an increase in λ increases the proportion of women who work and marry by increasing investment in market skills and by freeing more women from the household. Marriage probabilities by assumption remain unchanged, but the number of married women who work increases, whereas the number of married women who do not work falls.¹⁴ Thus, both models give rise to the same qualitative dynamics.

Depending on what we assume about payoffs at $\lambda = 0$, both models are capable of generating $\lambda = 0$ as a stable or unstable steady state, or not as a steady state at all. To obtain the latter, it is sufficient to assume that at $\lambda = 0$ it is still attractive to invest in a level of market skills such that at least some women end up working and married (and hence next period's $\lambda > 0$). In that case, the transition to an interior steady state is monotonically increasing from $\lambda = 0$ to λ^* , as shown in the upper λ_{t+1} curve in Figure I. Alternatively, if at $\lambda = 0$ no woman wishes to invest in market skills, and if this implies that no married woman works, then depending on the slope of the λ_{t+1} curve, one can have $\lambda = 0$ as a stable or unstable steady state.¹⁵

In terms of the historical evolution of women's participation in market work, one can perhaps think that the economy was slowly progressing along the top λ_{t+1} curve discussed previously, and that the advent of World War II with its attendant influx of mothers into market work, accelerated the transformation of woman's role in the economy. Alternatively, one may prefer to think that the economy was initially at a stable steady state at $\lambda = 0$, and that the growth of the service sector, or the diffusion of labor-saving household technology, or the decrease in the importance of the marriage bar, made it more attractive for at least some women to invest in market skills and work, thus shifting the λ_{t+1} curve from the bottom position in Figure I to the top one.¹⁶

We next turn to the empirical investigation of our hypothesis. We start with the cross-sectional evidence.

14. Alternatively, one could call all women who work less than some given number of hours "nonworking" women and those who work more "working" women, and the same dynamic analysis would go through.

15. Depending on functional forms, this model may be able to generate multiple steady states. This is not the focus of our analysis, however, so we do not explore this possibility further.

16. See Goldin [1990] for a discussion of the role of several of these factors in the historical evolution of women's labor force participation.

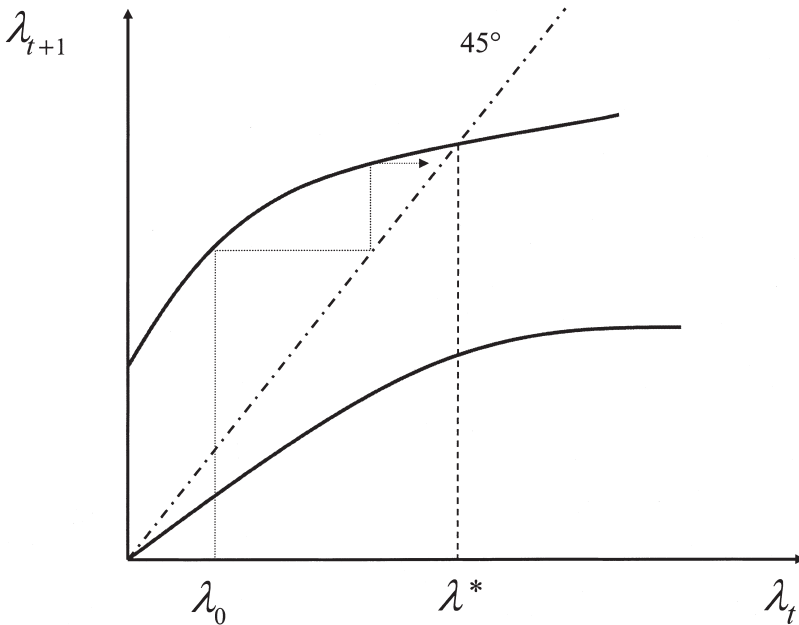


FIGURE I
Dynamics

II. CROSS-SECTIONAL EVIDENCE

In this section we present extensive evidence showing a positive correlation between the working behavior of a man's mother and that of his wife. In particular, we show that the working behavior of a man's mother has a large and significant impact on the likelihood that his wife works, even after controlling for several characteristics of husband and wife, and for various background characteristics of the couple (e.g., religion, geography, networks, etc.).

Our analysis is especially concerned with ruling out various background characteristics as the main drivers of our correlation. In particular, in order for the dynamic implications of our analysis to be correct (i.e., more mothers who work implying more women who work in the next generation), the positive correlation should not be driven by, say, assortative matching in religion (with some religions discouraging women from working more than others). If this were the main driver, then there would be no

dynamic implication to a shock that caused mothers to work more than they did previously.

We estimate the following model:

$$D_{it}^w = \beta_0 + \beta_1 X'_{it} + \beta_2 D_{it}^m + \varepsilon_{it},$$

where the dependent variable D_{it}^w is an indicator variable that captures the working decision of the wife, D_{it}^m is an indicator variable that is equal to 1 if the husband's mother worked while her son was growing up, and X_{it} is a vector of controls which varies with the particular specification considered.

Since no data set contains all the background information we are interested in for both spouses, we carry out our study using two different data sets that allow us to improve the analysis along different dimensions while corroborating the main result. We start with the General Social Survey (GSS). This data set contains not only information on the working behavior of the husband's mother—the main variable of interest in this analysis—but also on a number of other background characteristics of the husband, such as the type and region of residence where he grew up, the wealth of his family of origin, and the religion in which he was raised, which theoretically could be correlated not only with the working behavior of his mother but, through the matching process, could also influence the working behavior of his wife.

Although the GSS data set allows us to examine some competing hypotheses for our correlation, it has two shortcomings: the first is that when we include all the background variables, the number of observations is significantly reduced; the second is that the GSS lacks simultaneous information on the working behavior of both the husband's and wife's mothers. Information on the working behavior of the wife's mother is particularly important since the correlation between mother-in-law and wife could be driven primarily by “network” effects. That is, it could be that the working behavior of a husband's mother is correlated with that of his wife's, but that the primary channel driving the wife's working decision is the behavior of her own mother. It is primarily to examine this hypothesis that we turn to the “Female Labor Force Participation and Marital Instability” (FLFPMI) data set.¹⁷ This data set has the couple, rather than the individual, as the unit of analysis and contains background information for both husband and wife, even if less comprehensive than that found in the GSS

17. To our knowledge, in the past this data set has been used only by sociologists.

for the husband only. It also allows us to significantly increase the number of observations.

All the empirical evidence leads us to conclude that the working behavior of a woman is significantly and strongly correlated with the working behavior of her husband's mother. Depending on the data set of reference, *ceteris paribus*, a working mother increases the probability that a man's wife works by 24 to 32 percentage points.¹⁸

II.A. Working Behavior Analysis: The GSS

The GSS is a series of cross sections that have been collected annually since 1972 (except for a few years) by the National Opinion Research Center.¹⁹ Each cross section contains about 1500 observations, and respondents are asked about their demographic background, political and social attitudes, and labor market outcomes.

Our sample includes all white married men heads-of-households whose wives are between 30 and 50 years of age, as women in this age interval are more likely to have completed their education and are still far from retirement considerations.²⁰ The working behavior of the wife, our dependent variable in this analysis, is captured by an indicator variable (WIFWORK) that is equal to 1 if, during the week preceding the interview, she was employed full time or she had a regular job but was temporarily away from it because of illness, vacation, or strike and is equal to zero otherwise. The working behavior of the husband's mother, our variable of interest in this analysis, is described by a dummy variable (MAWORKH) that is set equal to one if the man's mother worked for as long as a year after her son was born and before he was fourteen, and zero otherwise.²¹ This variable is only available for the years 1988 and 1994. Hence we restrict our sample to these two years.

We control for several characteristics of the wife that may influence her working behavior such as her age and education, and for several characteristics of her husband that may influence her work choice, like his age, his years of completed education

18. We also carried out our analysis using the Panel Study of Income Dynamics (PSID), obtaining very similar results.

19. Davis, Smith, and Marsden [1999] describes the content and the sampling frame of the GSS.

20. We restrict our attention to whites as black women have had a very different history of labor force participation.

21. The correlation between a wife working and her mother-in-law working is 0.17, significant at the 1 percent level.

(HUSB EDUC), and his income (HUSB INCOME).²² We also add the number of children the husband has ever had (CHILDREN), and the number of children present in the household who are younger than age six (BABIES).

Our analysis includes a number of variables that capture other characteristics of the husband's background: in particular, his mother's and father's years of completed education (MAEDUCH and PAEDUCH), the religion in which he was raised (RELIGION 16: Protestant, Catholic, or Other) and whether he considers his family income at age sixteen to have been below average, average, or above average as compared with American families in general (INCOME 16). Finally, we include two variables that capture the location in which the husband lived at age sixteen: the first is a full set of dummy variables indicating the region in which he lived, and the second is a set of dummy variables indicating the type of place where he resided (respectively, RESIDENCE 16 and REGION 16).²³ The summary statistics for our sample are presented in Appendix 1.

Before turning to a discussion of our results, it is important to note that there does not appear to be any selection effect operating through the mother's work behavior on the probability that her son ends up married in the first place. Taking all men in the GSS between the age of 30 and 50 (over the two sample years mentioned previously), the raw probability that a man is married is 0.81 if his mother did not work and 0.86 if she did. Controlling for all his characteristics (age, education, income, etc.), including his background characteristics, the probability that he is married is not significantly affected by whether his mother worked: the coefficient on MAWORKH is .077 and insignificant.²⁴

Table I presents the results of our regressions. We report the

22. HUSB INCOME is labor earnings provided by the GSS in constant dollars (base = 1986) for the period 1974–1996. This variable is based on categorical data; income is calculated as the midpoint of the categorical variable. It is measured in thousands of dollars.

23. The regional variable is one of the following nine categories: New England (ME, VT, NH, MA, CT, RI), Middle Atlantic (NY, NJ, PA), East North Central (WI, IL, IN, MI, OH), West North Central (MN, IA, MO, ND, SD, NE, KS), South Atlantic (DE, MD, WV, VA, NC, SC, GA, FL, DC), East South Central (KY, TN, AL, MS), West South Central (AR, OK, LA, TX), Mountain (MT, ID, WY, NV, UT, CO, AZ, NM), and Pacific (WA, OR, CA, AK, HI). The residence variable is one of the following six categories: open country (but not on a farm), farm, small city, or town (under 50,000), medium-size city (50,000–250,000), suburb near large city and large city (over 250,000).

24. We performed the same exercise using the PSID and obtained similar results.

TABLE I
PROBIT WHETHER WIFE WORKS ON MOTHER'S WORKING BEHAVIOR (GSS)

	Marginal effects						
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
MAWORKH	.157** (.066)	.187*** (.063)	.165** (.067)	.211*** (.071)	.202** (.094)	.250*** (.093)	.323*** (.099)
HUSB AGE	.005 (.005)		.004 (.008)	.013 (.009)	.007 (.007)	.005 (.012)	.024* (.014)
HUSB EDUC	.013 (.013)		-.009 (.016)	-.027 (.016)	.00005 (.018)	-.049** (.023)	-.068*** (.024)
HUSB INCOME	-.007*** (.002)		-.007*** (.002)	-.006*** (.002)	-.008*** (.002)	-.008*** (.003)	-.008*** (.003)
WIFE AGE		.005 (.006)	.003 (.009)	-.008 (.010)		.0003 (.015)	-.021 (.017)
WIFE EDUC		.021 (.013)	.044*** (.016)	.052*** (.018)		.091*** (.028)	.100*** (.031)
CHILDREN				-.093*** (.032)			-.101** (.046)
BABIES				-.182*** (.062)			-.225*** (.078)
MAEDUCH					.009 (.020)	-.001 (.019)	-.011 (.024)
PAEDUCH					.007 (.017)	.007 (.017)	.019 (.020)
RELIGION					yes	yes	yes
INCOME 16					yes	yes	yes
RESIDENCE 16					yes	yes	yes
REGION 16					yes	yes	yes
N. obs.	231	251	230	229	160	160	159
Pseudo R^2	.062	.034	.084	.162	.198	.244	.329
Log/likelihood	-149.03	-166.82	-145.03	-132.17	-88.30	-83.13	-73.43

Marginal effects are calculated at the means of the independent variables. WIFEWORK = 1 if wife's employed full time, or temporarily away from job because of illness, vacation, or strike during the week preceding the interview. MAWORK = 1 if husband's mother ever worked for pay for as long as one year after he was born and before he was fourteen. RELIGION is a set of three dummies for religion in which husband was raised, INCOME is a set of three dummies for husband's self-assessment of family income at age sixteen. RESIDENCE is a set of six dummies for husband's type of residence at age sixteen and REGION is a set of nine dummies for the geographical region of husband's residence at age sixteen. Robust standard errors are in parentheses. All regressions include a constant term. * Significance at 10 percent level. ** Significance at 5 percent level. *** Significance at 1 percent level.

marginal effect of each variable.²⁵ Regression (i) in Table I is our baseline regression; it estimates the effect of the working behavior of the husband's mother on the probability that the wife

25. All the probit regressions in Table I are run including a constant and year-fixed effects and estimated using robust standard errors. For expositional purposes the coefficient on the constant term and the year effects are not reported in the table. In the regressions, the omitted variables are OTHER for the husband's religion dummies, BELOW for his self-assessment of family income at age sixteen, COUNTRY for his place of residency at sixteen, and East South Central for the regional dummies.

works, controlling only for the husband's characteristics of age, education, and income. The next specification uses only the wife's characteristics. Regression (iii) add the husband's characteristics to those of the wife, and specification (iv) includes the number of children and babies of the couple. We find that the probability that the wife works is positively and significantly related to whether her husband's mother has worked for all specifications. Also significant, as found in previous studies, are the number of children and babies in the household (negative effect), the wife's own education (positive effect) and her husband's income (negative effect).

Regression (v) presents the results obtained for model (i) augmented by all the husband's background variables, which include his parents' education, the religion in which he was raised, a self-assessment of his family financial situation at age sixteen, and the two sets of dummies capturing the husband's residence at age sixteen. The next specification once again includes the wife's characteristics, and the last specification reintroduces children and babies along with all the other variables. These specifications allow us to distinguish our hypothesis from different explanations according to which other background factors, such as geography, religion, and family wealth, may be driving the correlation between the working behavior of a man's mother and that of his wife. Below, we discuss the potential role these variables could play in greater detail.

One may argue that the observed positive correlation in work behavior simply reflects that a man whose mother worked is more likely to have been raised in an area where women are more likely to work. If he also married a woman from that area, this would then produce the positive relationship. In particular, one might think that women who live in cities are more likely to work, either because of social norms, self-selection, or greater opportunity. To control for this possibility, we include both residence and region dummies. Similar reasoning also leads us to control for religion. People tend to marry others of the same religion. As discussed previously, if some religions systematically encourage women to work, then this may be responsible for the positive coefficient on MAWORKH.

Lastly, it may be argued that a man's mother is more likely to have worked if her husband's income or their joint wealth was low. Since a man coming from a low-income family is more likely to have low family wealth himself (via bequests or other channels

of wealth persistence), his wife would also be more likely to work, even after controlling for his (annual) income. In this case, the positive coefficient on MAWORKH may simply be picking up the negative correlation between family wealth and the wife's working behavior.²⁶

Controlling for all the aforementioned background variables only increases the coefficient on MAWORKH. As shown in the last column of Table I which includes all our controls, we find that having a husband whose mother worked increases the probability that a married woman works full time by 32 percentage points, from 39 percent to 71 percent, and the effect is significant at the 1 percent level. This is a very large effect. Note that, for example, the presence of an additional baby reduces the probability that a wife works full time by about 22 percentage points, whereas an additional year of her own education increases this probability by about 10 percentage points.

All the results are robust to alternative definitions of the dependent variable: whether we define a wife as working when she works full time or part time, or when she works more than 40 hours per week, we obtain similar results.²⁷ Adding squared terms for the husband's age, the wife's age, and the husband's income also leaves our results unchanged. On the other hand, if we use a different indicator of the husband's mother's working history, such as whether she worked for as long as a year at any point after marriage, the results no longer hold, and the coefficient on the working behavior of the husband's mother becomes insignificant. This most likely results from the fact that most mothers have worked for at least a year at some point during their married life. A last alternative, whether the man's mother worked after he was born and before he started first grade, also has the mother's working behavior entering positively and significantly in determining the probability that the son's wife works. Thus, what seems to matter is whether a man's mother worked while he was relatively young.

We also explored the possibility that whether a married woman works depends on whether her mother-in-law herself

26. Since our data set does not have any information on the husband's parental wealth or income, we included in the regression a set of dummy variables which provide a measure of the parents' income/wealth (compared with that of the average American family) as assessed by the son.

27. The marginal effect of MAWORKH decreases to 20 percentage points if we include "part time" in our definition of wifework but the coefficient remains significant at the 1 percent level.

worked in a more or less prestigious job. As we do not have the data that allow us to examine the mother-in-law's job prestige, we attempted to capture this element by introducing an interaction term that equals one if both MAWORKH is equal to one and MAEDUCH is greater than 12 (i.e., the husband's mother has more than high school diploma), and is equal to zero otherwise. This term was negative and insignificant in all the models estimated in Table I. The coefficient on MAWORKH increased slightly in all specifications, and the marginal effect in the full specification increased to 35 percentage points. Thus, it does not appear that the possible stigma that may exist from having a mother work in what is probably a less prestigious job affects the strength of the transmission mechanism from mother to son, at least not during the time period of our analysis.²⁸

To summarize, the GSS allowed us to distinguish our hypothesis from several competing explanations. As mentioned previously, however, it does not allow us to control for the working behavior of the wife's own mother. To examine whether our results are robust to the addition of variables that capture the wife's background that are not available in the GSS, we next turn to the FLFPMI.²⁹

II.B. Working Behavior Analysis: The FLFPMI

This data set consists of a national probability sample of 2,034 married men and women under 55 years old who were interviewed by telephone in the fall of 1980. The data collection was designed to study the effect of wives' participation in the labor force on marital instability and includes information on working behavior, earnings and occupational status as well as detailed background variables for both spouses [Booth, Alan, et al. 1980]. Particularly important for our analysis, this data set has the rare characteristic of including retrospective information on the working behavior of the mothers of both spouses. The characteristics of the sample were compared with estimates made by the U. S. Census Bureau, and the sample was found to be nationally representative with respect to age, race, household size, presence of children, region, and female labor force partici-

28. We thank an anonymous referee for bringing this possibility to our attention.

29. The only background variables available in the GSS for the wife are the education of her parents and the religion in which she was raised. However, the information on the education of her parents is only available in 1988 so it greatly reduces the number of observations. Including the religion in which she was raised leaves the results unaltered.

pation. The sample was weighted to take into account a slight underrepresentation of people in metropolitan areas.

As in the previous analysis, we restrict the sample to include all white couples. Some variables have a slightly different definition than in the GSS. The working behavior of the wife, WIFEWORK, is described by a dummy variable that is equal to one if, at the time of the interview, the wife was working for pay in a full-time job and is equal to zero otherwise. The working behavior of her husband's mother, MAWORKH, on the other hand, is now captured by a dummy variable that is set equal to one if the man's mother worked "all the time" while her son was growing up, and zero otherwise.³⁰

As in the previous analysis we control for several characteristics of husband and wife that may influence the wife's working behavior such as age, years of completed education, husband's income, and number of children.³¹ Differently from the previous analysis, we are also now able to include variables that capture some background characteristics of the wife: in particular, we have information on the working behavior of the wife's own mother while she was growing up, which is described by the variable MAWORKW that, symmetrically to MAWORKH, is set equal to one if the wife's mother worked "all the time" while her daughter was growing up. In general, we indicate the wife's variables with a W and the husband's with an H. This data set also contains the years of completed education of mother and father for both spouses, their religion (RELIG: Protestant, Catholic, None, or Other) and the Duncan socioeconomic index for the father of both spouses (SOCECPA) which is meant to capture the father's occupational prestige and is used here as a proxy for the financial situation of the family of origin.³²

Since the FLFPMI does not have information on the type of place and geographic region in which the two spouses grew up, we include in the analysis two variables that describe the location of

30. The data set provides five categories for the past working behavior of the husband's mother. In addition to working "all the time" while her son was growing up, there is also "most of the time," "about half," less "than half" and "never."

31. HUSB INCOME is calculated as the husband's percentage contribution to family income times family income in thousands of dollars. Family income is provided in ten categories and calculated as the midpoint of the categorical variable. The last category has been adjusted for top coding by multiplying by 1.2.

32. The Duncan socioeconomic index is a measure of occupational status based upon the income level and educational attainment associated with each occupation in 1950. The score was derived by using median income and education levels for men in 1950 to predict prestige assessments from a 1947 survey (of a selected group of occupations). The resulting statistical model was used to generate scores for the entire range of 1950 occupations. See Duncan [1961].

residence of the couple at the time of the interview: the first, *REGION*, is a full set of dummy variables indicating the region in which they live, and the second, *RESIDENCE*, is a set of dummy variables indicating the type of place where they reside.³³

The summary statistics for our sample are presented in Appendix 1 and are very similar to those obtained for GSS. The only substantial difference is that now the percentage of men who have had a working mother is lower than in the GSS sample, reflecting the different definition of the variable used to capture the working behavior of the husband's mother in this data set. The sample size is about five times greater than for the GSS.

Table II reports the results of our regression analysis. The first five specifications are basically the same as in Table I. Our results are similar to the ones we found for the GSS. In specification (vi) we control for the working behavior of the wife's mother. The two *MAWORK* variables have a correlation of approximately zero (0.05) suggesting that a "network" effect is not at work. This is corroborated since we find that *MAWORKH* remains significant at the 1 percent level and that its coefficient increases in magnitude when we include *MAWORKW*. Perhaps more surprising is that the working behavior of the wife's mother is not significant in explaining the wife's working behavior. In any case, we can now reject the possibility that our positive correlation simply reflects assortative matching. It does not appear to be true that men whose mothers worked marry women whose mothers also worked and that this is what lies behind our positive correlation. The next specification leads to similar results.

Finally, column (vii) presents the results obtained for model (vi) augmented by the characteristics that are common to the couple: number of children, geographic region, and type of residence.³⁴ Once again, after including all controls, we find a large, positive, and significant effect of a mother-in-law who worked on the probability that a wife works. The probability that a wife

33. The regional variable has been constructed from telephone area codes and consists of the following nine categories: New England (ME, VT, NH, MA, CT, RI), Middle Atlantic (NY, NJ, PA), East North Central (WI, IL, IN, MI, OH), West North Central (MN, IA, MO, ND, SD, NE, KS), South Atlantic (DE, MD, WV, VA, NC, SC, GA, FL, DC), East South Central (KY, TN, AL, MS), West South Central (AR, OK, LA, TX), Mountain (MT, ID, WY, NV, UT, CO, AZ, NM) and Pacific (WA, OR, CA, AK, HI). The residence variable consists of the following three categories: open country, farm, and town or city.

34. In the regressions the omitted variables are *OTHER* for the religion dummies, *FARM* for the type of residence, and *NEW ENGLAND* for the regional dummies.

TABLE II
PROBIT WHETHER WIFE WORKS ON MOTHER'S WORKING BEHAVIOR (FLFPMI)

	Marginal effects							
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
MAWORKH	.093** (.041)	.103*** (.039)	.091** (.042)	.089** (.042)	.092* (.048)	.164*** (.052)	.211*** (.063)	.241*** (.064)
HUSB AGE	.004** (.001)		.008** (.004)	.011*** (.004)	.006*** (.002)		.001 (.008)	.007 (.008)
HUSB EDUC	-.002 (.005)		-.024*** (.006)	-.027*** (.006)	-.001 (.007)		-.034*** (.011)	-.042*** (.012)
HUSB INCOME	-.012*** (.001)		-.013*** (.001)	-.012*** (.001)	-.013*** (.001)		-.011*** (.002)	-.011*** (.002)
WIFE AGE		-.002 (.001)	-.005 (.004)	-.002 (.004)		-.002 (.002)	.004 (.008)	.010 (.008)
WIFE EDUC		.013** (.006)	.048*** (.008)	.042*** (.008)		.016* (.009)	.049*** (.014)	.039*** (.014)
CHILDREN				-.065*** (.011)				-.094*** (.022)
MAEDUCH					.003 (.007)		.011 (.011)	.008 (.011)
PAEDUCH					.001 (.006)		-.019* (.010)	-.020* (.011)
PASOCECH					.0003 (.0007)		.002 (.001)	.002 (.001)
MAWORKW						.001 (.064)	-.056 (.082)	-.066 (.085)
MAEDUCW						.006 (.008)	.007 (.010)	.013 (.011)
PAEDUCW						-.009 (.007)	-.0002 (.009)	.0004 (.010)
PASOCECW						-.002** (.001)	-.002 (.001)	-.002 (.001)
RELIGH					yes		yes	yes
RELIGW						yes	yes	yes
RESIDENCE								yes
REGION								yes
N. obs.	1454	1535	1453	1449	1072	796	530	528
Pseudo R^2	.060	.007	.081	.099	.062	.026	.106	.107
Log/likelihood	-943.25	-1052.14	-921.47	-902.08	-692.92	-536.43	-328.31	-308.80

Marginal effects calculated at the means of the independent variables. WIFEWORK = 1 if, at the time of the interview, the wife was working for pay in a full-time job. MAWORK = 1 if the husband's mother worked all the time while her son was growing up. PASOCEC is the Duncan socioeconomic index of the father. RELIG is a set of four religion dummies, RESIDENCE is a set of three dummies for the type of place where the couple reside, REGION is a set of nine dummies for the geographical region where the couple reside. Robust standard errors are in parentheses. All regressions include a constant term. * Significance at 10 percent level. ** Significance at 5 percent level. *** Significance at 1 percent level.

works increases by 24 percentage points, from 46 percent to 70 percent.

As before, our results are robust to alternative definitions of the dependent variable: whether we define a wife as working

when she works full time or when she just works for pay, we obtain similar results.³⁵ Adding squared terms for the husband's age, the wife's age, and the husband's income also leaves our results unchanged. If we use as an indicator of the husband's mother's working history not whether she worked "all the time" while her son was growing up, but instead whether she worked "most of the time," the mother's working behavior still enters positively and significantly in determining the probability that the son's wife works, but its marginal effect is about 11 percentage points.

Our results show that whether a man's mother worked has a strong effect on whether his wife works. Before turning to the dynamic empirical analysis, it is useful to dispel one concern. An alternative hypothesis to the one that we entertain is that the correlation we document results from the transmission of some gene that makes a woman more predisposed both to work and to have sons who like working women. If this were the case, the dynamics of our model would no longer hold.³⁶ The number of working mothers might increase for a variety of reasons, but this would not make it more attractive for women from the next generation to work since there would not be a corresponding change in the pool of men (as this is governed by genetics, and not by the mother's working behavior *per se*). Note, however, that this alternative hypothesis implies that the correlation between men whose mother worked and men with working wives should decrease over time. This would happen because the greater the proportion of women who work, the less likely it is that having a working mother is correlated with a man's (genetic) predisposition to favor working women. Surprisingly (since, with the diffusion of generally more favorable attitudes toward working women, one would expect working wives to become more common for all types of men), our GSS sample shows that the correlation between the working behavior of a man's wife and that of his mother actually increased over time.³⁷

We next turn to the intergenerational evidence.

35. The marginal effect of MAWORKH decreases to 17 percentage points using this looser definition but remains significant at the 1 percent level.

36. We thank Lawrence Katz and an anonymous referee for bringing this potential problem to our attention.

37. We split our sample into two time periods. The first cohort consists of men born 1940–1953; the second is of men born 1954–1966. The correlation between mother-in-law and wife for the first cohort is .12 (significant at the 15 percent level); that of the second cohort is .24 (significant at the 5 percent level).

III. INTERGENERATIONAL EVIDENCE

According to our theory, an event that increases the proportion of men brought up by working mothers has dynamic repercussions on women's labor supply since it makes work more attractive to women in the next generation. In this section we examine whether this intergenerational mechanism is at work by exploring the dynamic consequences of two different sources of variation.

We first make use of variation across U. S. states in the importance of World War II as a shock to female labor supply, to provide exogenous variation in the proportion of men raised by working mothers. We contrast the consequences of this variation on a cohort that was too young to be directly affected by the war, but who would be affected by a change in the proportion of boys brought up by working mothers, with that of cohorts who were directly affected by the war but were too old to be affected by their mother's working behavior.

Our second source of variation comes from differences across the United States in the proportion of men raised by working mothers relative to those raised by nonworking mothers. Our theory implies that, for a given level of female labor supply, states with higher ratios of the average fertility of working relative to nonworking women should have greater female labor supply the following generation. We examine the relationship between this relative fertility ratio and next generation's female labor supply across the United States over a 30-year period.

III.A. *World War II*

World War II can be usefully viewed as providing an "exogenous" shock to female labor supply. As men were mobilized to serve in the war, women increased their labor force participation markedly. In 1940 only 28 percent of women over age fifteen participated in the labor force. By 1945 this figure exceeded 34 percent.³⁸

Acemoglu, Autor, and Lyle [2004] argue that variations in the importance of this shock across states—captured by differentials in the mobilization rate of men across states—can be used to provide exogenous variation in women's labor supply. Their em-

38. See Blau, Ferber, and Winkler [2000]. Goldin [1991] shows that half of the women who entered the labor force during the war period had exited the labor force by 1950, still leaving a large increase in participation.

pirical strategy is to use World War II mobilization rates as the first-stage regression in an analysis of the effects of the increase in female labor supply on the wage structure. In this paper we also make use of the variation provided by differences in mobilization rates on female labor supply across states. Unlike these authors, however, we are interested in identifying the effects of this variation on the labor supply of women many years later, and most importantly, we wish to identify the “echo” effect that this variation should have, according to our theory, for the cohort of women who were young enough during World War II to be affected by the change in the available pool of men in the next generation.

The basic logic of our exercise is as follows. World War II directly affected the work behavior of women during the war years. As we will show, the differential effect of the war did not fade immediately; rather it lingered for several decades in the work behavior of those women who were old enough in the 1940s to be directly affected by the war. As these women aged, however, the effect of the war on their work behavior appears to have slowly faded. By the time these women reached the age of 45–50, there does not appear to be a differential effect of the war on their labor supply. A younger generation of women—those born in 1930–1935 (who were thus 7–12 years old in 1942)—was too young to be directly affected by the war, but not too young to be affected by the change in their mothers’ work behavior. As we show, the war affected this cohort’s labor supply as well. Most importantly, although the effect of World War II faded for the older cohorts, its influence on the labor supply of this later cohort persisted. This is an effect that our theory would predict, as the change in these women’s labor supply did not depend on whether they worked during the war, but rather on the expectations they formed, then and after, as to the return of investing in market skills. The return to becoming a working woman had increased, according to our hypothesis, since more boys had been raised by a working mother.

We investigate our hypothesis in several steps. The first step is to show that, as posited, the mobilization rate of men during World War II had a positive effect on the labor supply of the mothers of the 1930–1935 cohort. The second step is to trace out the echo effect of the war over the life cycle of the 1930–1935 cohort by contrasting the indirect effect of the war on the labor

supply of this cohort with the direct effect of the war on older cohorts at the same age.

III.B. The Data Set

We use data from the 1 percent Integrated Public Use Microsample (IPUMS) of the decennial Census for the decades 1940 to 1980.³⁹ We restrict our attention to white married women belonging to the following three age groups: 25–30, 35–40, and 45–50. We exclude women living on farms or working in agricultural occupations, as well as those living in group quarters (e.g., prisons, and other group living arrangements such as rooming houses and military barracks).⁴⁰

Our primary measure of female labor supply is the number of weeks worked in the previous year. In 1960 and 1970, Census information on weeks worked is reported in intervals (1–13 weeks, 14–26 weeks, 27–39 weeks, 40–47 weeks, 48–49 weeks, and 50–52 weeks). For these decades we compute our measure of weeks worked by assigning the midpoint of each interval. For 1940, 1950, and 1980 we use the information on actual number of weeks worked that is available in the Census.⁴¹ For 1950, information on weeks worked is only available for sample line persons; hence we use the appropriate sample line weights in the analysis. For the remaining decades we use the appropriate person weights that indicate the number of people in the population that each sampled individual represents. The summary statistics for our sample are reported in Appendix 2. Since we assign mobilization rates by women's state of birth, we exclude women born outside the United States as well as those born in Alaska and Hawaii since these were not states until the 1950s.

Our mobilization rate variable is the same used in Acemoglu, Autor, and Lyle [2004].⁴² They use published tables from the

39. In particular, we use the general 1 percent sample for 1940, 1950, and 1960. For the 1970 we use the 1 percent State Sample (Form 1), and for the 1980 we use the 1 percent Metro Sample (Sample B).

40. We exclude the following occupations (based on the 1950 Census definition): farmers (owners and tenants), farm managers, farm foremen, farm laborers as wage workers, farm laborers as unpaid family workers, and farm service laborers as self-employed.

41. In the 1940 Census respondents were required to report this information in terms of "equivalent full-time weeks." It was up to respondents to determine precisely what "full-time" meant, though enumerators were instructed to suggest that 40 hours was a good round figure. In essence, respondents were to estimate how many hours they had averaged per week, multiply this figure by 52 weeks, then divide by 40 (See Census codebook).

42. We thank the authors for making the data available to us.

Selective Service System [1956] and construct men's mobilization rates during World War II as the fraction of the 18 to 44 years old registered males in a state who were drafted for war.⁴³ The average mobilization rate was .474 with a standard deviation of .035.

Mobilization rates varied substantially across states, from less than 42 percent in Georgia, the Dakotas, and the Carolinas, to more than 52 percent in Washington, Pennsylvania, New Hampshire, Oregon, and Massachusetts. The state differences in war mobilization reflect a variety of factors. The Selective Service's guidelines for deferments were based on marital status, fatherhood, essential skills for civilian war production, and temporary medical disabilities, but also left considerable discretion to the local boards. Farm employment, in particular, was a major cause of deferment as maintaining food supply was considered essential to the war effort, and, not surprisingly, states with a higher proportion of men who are farmers have a lower mobilization rate. The mobilization rate is also higher in states with higher average male education and with a lower percentage of black males.

To attempt to control for systematic variation in the mobilization rate, our regressions include the 1940 fraction of nonwhite men aged 13 to 44, the 1940 fraction of men between the ages of 13 to 44 who are not farmers, and the 1940 average education of men in this same age group.⁴⁴ As shown in Acemoglu, Autor, and Lyle [2004], after controlling for these factors and for other non-economic components (such as the age composition and the number of German-born men), there is still some 30 percent variation of mobilization rates across states that is left unexplained and which is attributed to idiosyncratic strategies followed by local registration boards.⁴⁵

III.C. Model Specification and Results

We now describe our analysis of the impact of World War II on the working behavior of married women. Our analysis pools married women born in 1930–1935 with married women from

43. Since all men in the age bracket 18–44 were registered, their mobilization rate variable represents the fraction of men in this age range who served. Mobilization rates for Nevada and Washington, DC are not available (the former because it saw large population changes during this time period).

44. Men who were 13 in 1940 would be 18 in 1945 and therefore part of the draft target group.

45. See Table 4 in Acemoglu, Autor, and Lyle [2004].

earlier cohorts and contrasts the indirect effect of the war on the 1930–1935 cohort with the direct effect of the war on the earlier cohorts. For a meaningful comparison, we look across cohorts at the same age (i.e., we take their labor supply in the decade in which they reach the specified age).

We run the following basic regression:

$$(11) \quad w_{ist} = X'_{ist}\beta_1 + \gamma_t \tilde{X}'_{ist}\beta_{2t} + \gamma_t e'_s\beta_{3t} + \alpha_t \gamma_t m_s + d_s + \gamma_t + \varepsilon_{ist},$$

where w_{ist} measures weeks worked by woman i born in state s at time t , X'_{ist} represents a set of individual characteristics: age dummies, state of residence dummies, and husbands' state of birth dummy. In $\tilde{X}'_{ist}$ the age dummies are interacted with a year effect γ_t (for each decade following 1940). We also include a year dummy, a state dummy d_s , and the aforementioned set of state-level 1940 economic variables (farmers, nonwhites, and average education) interacted with the year dummy.

Our variable of interest is the interaction of the mobilization rate variable m_s assigned by female state of birth, with the time dummy γ_t . The coefficient α_t measures whether states with higher mobilization rates during World War II experienced a larger increase in female labor supply in decade t . Since the key variable on the right-hand side only varies by state and year, all the standard errors we report in this experiment are corrected for clustering at the state-year level.

In 1935 the average age of a white woman giving birth was 26.8.⁴⁶ Thus, we study the mothers of our cohort of interest (1930–1935) by examining the labor supply of married women born in 1903–1908. We run a regression similar to the one specified in (11), pooling the observations for the mothers' cohort with that of the cohort born ten years earlier. Instead of assigning variables by state of birth, however, we assign them by state of residence since this is more likely to correspond to the state where the woman lived during World War II. As the 1903–1908 cohort was 42 to 47 years old in 1950, we pool 42–47 years old in 1940 with the same age group in 1950. Table III reports the results.

As shown in the table, we obtain a point estimate for α of

46. Calculated from the Statistical Tables on Births: Live Births by Age of Mother and Race: United States, 1933–1998 (National Center for Health Statistics web page).

TABLE III
IMPACT OF WORLD WAR II ON LABOR SUPPLY OF 42–47 YEAR-OLD MARRIED WOMEN.

Dependent variable is “Weeks Worked”	
	1940 & 1950
1940 mobilization rate $\times$ year	19.64** (8.71)
1940 share male nonwhite $\times$ year	–2.37 (4.66)
1940 share male farmer $\times$ year	6.12*** (1.79)
1940 male avg years educ $\times$ year	.218 (.483)
Year	–6.80 (6.01)
N. obs.	32,347
Adjusted R^2	0.03

Robust standard errors in parentheses account for clustering at the state-year level. Estimation results are for a regression that pools together 42 to 47 year-old women across two cohorts: the 1903–1908 cohort in 1950 and the 1893–1898 cohort in 1940. The dependent variable, weeks worked, is regressed on the mobilization rate variable (interacted with a 1950 dummy) assigned by the woman’s state of residence. We also control for state fraction of male farmers, the state fraction of nonwhite males, and the state average education of males in 1940. All the 1940 variables (interacted with the 1950 dummy) are assigned by the woman’s state of residence. All specifications include state of residence dummies, a 1950 year dummy, age dummies, and the latter interacted with a 1950 dummy. Data are from Census IPUMS 1 percent sample for both years. For 1950 they refer to the sample online subsample. The regression is weighted by census sampling weights. Our sample consists of 42 to 47 year-old white married women residing in the mainland United States excluding Nevada and DC, not living in institutional quarters, not living on farms or working in agricultural occupations. * Significance at 10 percent level. ** Significance at 5 percent level. *** Significance at 1 percent level.

19.6. This implies that a 10 percent increase in the mobilization rate is associated with an increase in female employment for this age group of 1.96 weeks between the beginning and the end of the 1940 decade. To interpret the magnitude of this effect, note that white married women of this age were working on average 6.7 weeks in 1940. Hence this number represents an increase of around 30 percent in their labor supply. Thus, we conclude that the women most likely to have children in 1930–1935 significantly increased their labor supply substantially more in those states in which the mobilization rate of men was greater. These states, therefore, would have had a greater increase in the proportion of men brought up by working mothers.

We now turn to the analysis of the labor supply of our cohort of interest: women born in 1930–1935. We proceed similarly to what we did for the mothers’ cohort, by pooling the 1930–1935 cohort with preceding ones at a given point in their life cycle, but

assigning variables by state of birth. Our results are reported in Table IV.

We start by examining cohorts at the age of 25–30 (and thus pool our cohort in 1960 with the 1920–1925 cohort in 1950 and the 1910–1915 cohort in 1940). As shown in column (i) of panel A, the coefficient on the mobilization rate interacted with 1950 is positive (and almost significant at the 10 percent level). We would interpret this as the direct effect of the war on the women born in 1920–1925, as they were close to their early twenties during the war. What we call the “indirect” effect of the war can be seen in the coefficient on the mobilization rate interacted with 1960. The 1930–1935 cohort was too young to be affected directly by the war, but not too young to be affected by the fact that they had more working mothers. As our theory predicts, the coefficient on the mobilization rate in 1960 is positive and significant, showing that this cohort was also affected, albeit indirectly, by the war.

The next two columns in panel A of Table IV repeat this regression with some modifications. The second column includes dummies for the state of residence and for the husband’s state of birth. Of course, the state of birth of a woman’s husband is an endogenous variable (in the sense that she chooses whom to marry and this may be a relevant characteristic), as is her state of residence. The question is whether a woman born in a particular state, with a given mobilization rate, formed her expectations about the proportion of men with working mothers in that state, or in her current state of residence. This depends on when she moved, among other things. As this is something we cannot determine, we find it of interest to run our regression both including and omitting these controls.

The third column uses the same variables as in the preceding specification, but restricts the sample to those women whose state of residence is the same as their state of birth. This allows us to not worry about what should be the “correct” mobilization rate for women who moved. In all regression specifications, the mobilization rate is positive (and almost significant) in 1950, and it is positive and significant in 1960. Note that the coefficient of 26 implies that a 10 percent increase in the mobilization rate increased this age group’s labor supply by 2.6 weeks. As this age group was working 8.83 weeks in 1940 and 11.78 weeks in 1960, this would represent 90 percent of the total increase in weeks worked over this time period.

Panels B and C in Table IV repeat the same exercises as

TABLE IV
IMPACT OF WORLD WAR II MOBILIZATION RATES ON LABOR SUPPLY OF MARRIED WOMEN

Dependent variable is "Weeks Worked"				
Panel A: 25–30				
	(i)	(ii)	(iii)	(iv)
1940 mobilization rate $\times$ 1950	18.11 (11.05)	17.29 (10.99)	21.68 (14.50)	22.59 (14.24)
1940 mobilization rate $\times$ 1960	22.71** (11.30)	19.06* (11.12)	26.68* (15.1)	26.39* (15.15)
Year 1950	-11.23* (6.26)	-11.49* (6.16)	-19.44** (7.51)	-17.06** (8.19)
Year 1960	-5.58 (5.89)	-3.24 (5.72)	-11.73* (7.02)	-14.54* (7.47)
St. of residence & husband's st. of birth		yes	yes	yes
Education				yes
N. obs.	75,748	73,710	50,146	50,146
Adjusted R^2	0.01	0.015	0.016	0.027
Panel B: 35–40				
	(i)	(ii)	(iii)	(iv)
1940 mobilization rate $\times$ 1950	25.25*** (8.36)	23.67*** (7.92)	33.02*** (9.99)	34.49*** (10.28)
1940 mobilization rate $\times$ 1960	18.34*** (6.76)	18.17*** (6.29)	22.55*** (7.89)	24.74*** (8.22)
1940 mobilization rate $\times$ 1970	14.24* (8.07)	14.78** (7.51)	22.01** (8.74)	25.12*** (8.64)
Year 1950	-2.25 (5.87)	-1.12 (5.63)	-11.88 (7.79)	-20.55*** (7.90)
Year 1960	-3.76 (4.91)	-3.12 (4.75)	-7.30 (6.73)	-16.73*** (6.81)
Year 1970	1.79 (6.02)	.99 (5.65)	-6.19 (7.87)	-7.58 (9.02)
St. of residence & husband's st. of birth		yes	yes	yes
Education				yes
N. obs.	112,125	109,864	71,018	71,018
Adjusted R^2	0.039	0.045	0.05	0.05

above for the ages of 35–40 and 45–50. As before, the first column does not include dummies for the state of residence and the husband's state of birth, and the third column is restricted to women who reside in the same state as their state of birth.

Panel B shows that both the 1910–1915 and the 1920–1925

TABLE IV
(CONTINUED)

Dependent variable is "Weeks Worked"				
Panel C: 45–50				
	(i)	(ii)	(iii)	(iv)
1940 mobilization rate $\times$ 1950	12.59 (11.39)	17.92 (11.59)	27.98* (14.79)	26.17* (14.79)
1940 mobilization rate $\times$ 1960	11.47 (9.24)	16.22* (9.29)	15.44 (12.54)	15.33 (12.59)
1940 mobilization rate $\times$ 1970	3.25 (8.56)	8.99 (8.75)	13.77 (12.31)	15.96 (12.23)
1940 mobilization rate $\times$ 1980	17.26* (10.18)	21.72** (10.76)	32.89** (14.78)	33.07** (14.78)
Year 1950	−14.55 (9.59)	−17.96** (9.44)	−27.31** (13.21)	−26.23** (13.08)
Year 1960	.52 (6.53)	−2.09 (6.44)	−7.95 (10.39)	−11.71 (10.81)
Year 1970	9.50 (6.84)	5.07 (6.79)	1.82 (10.27)	−8.88 (10.59)
Year 1980	8.18 (7.46)	5.01 (7.60)	−1.84 (11.39)	−7.01 (11.80)
St. of residence & husband's st. of birth		yes	yes	yes
Education				yes
N. obs.	129,899	126,715	80,261	80,261
Adjusted R^2	0.087	0.091	0.098	0.11

Robust standard errors in parentheses account for clustering at the state-year level. Panel A pools 25–30 year-old women across cohorts by taking the 1930–1935 cohort in 1960, the 1920–1925 cohort in 1950, and the 1910–1915 cohort in 1940; panel B pools 35–40 year-old women across cohorts by taking the 1930–1935 cohort in 1970, the 1920–1925 cohort in 1960, the 1910–1915 cohort in 1950, and the 1900–1905 cohort in 1940; panel C pools 45–50 year-old women across cohorts by taking the 1930–1935 cohort in 1980, the 1920–1925 cohort in 1970, the 1910–1915 cohort in 1960, the 1900–1905 cohort in 1950, and the 1890–1895 cohort in 1940. The dependent variable, weeks worked, is regressed on the mobilization rate variable (interacted with year dummies) assigned by the woman's state of birth. We also control for state fraction of male farmer, the state fraction of nonwhite male, and the state average education of males in 1940. All the 1940 variables (interacted with year dummies) are also assigned by the woman's state of birth. All specifications include state of birth dummies, a year dummy, age dummies, and the latter interacted with a year dummy. Specification (ii) also includes state of residence dummies and husband's state of birth dummies. Specification (iii) restricts the sample to women who were born in the same state they reside in. Specification (iv) includes eight education dummies and their interaction with year dummies. Education is measured as the highest grade of school attended or completed by the respondent. Data are from Census IPUMS 1 percent sample for all years. For 1950 they refer to the sample line subsample. All specifications are weighted by census sampling weights. Our sample consists of the age groups we study for white married women born in mainland United States excluding Nevada and DC, not living in institutional quarters, not living on farms or working in agricultural occupations. * Significance at 10 percent level. ** Significance at 5 percent level. *** Significance at 1 percent level.

cohorts worked more at the age of 35–40 (i.e., in 1950 and 1960, respectively) in states that had higher mobilization rates. This represents the direct effect of the war on these cohorts. As shown

by the third entry in the columns, the indirect effect of the war is also present: the 1930–1935 cohort also worked more in 1970 in states with higher mobilization rates. It is interesting to note that already at this point we can see the direct effect of the war fading as shown in the decrease in the coefficient that accompanies the mobilization rate from 1950 to 1960. The coefficient on the mobilization rate in 1970 in the third column implies that an increase in the mobilization rate of 10 percent accompanied a 2.2 week increase in the labor supply of women of age 35–40 in 1970. As weeks worked by women of this age increased from 7.44 weeks in 1940 to 18.02 weeks in 1970, this would represent almost 21 percent of the total increase in weeks worked over this time period.

Lastly, panel C examines women at the age of 45–50. As by the time women reach this age it is more likely they no longer reside in their state of birth (indeed, our sample of women decreases by a third), we will concentrate on the results reported in the third column. As shown, there was a direct effect of the war on women from the 1900–1905 cohort in 1950. However, unlike the other cases, the effect of the war has basically completely worn off by the time the next two cohorts (1910–1915 and 1920–1925) reach 45 to 50, in 1960 and 1970, respectively. The coefficient on the mobilization rate is insignificant. The result is dramatically different for our 1930–1935 cohort: in 1980 the effect of the war on women is again positive, statistically significant, and quantitatively important. Women of this age group worked around 3.3 weeks more in 1980 than in 1940 in states with a 10 percent higher mobilization rate. Since weeks worked by women in this age group increased from 5.5 weeks in 1940 to 26.26 weeks in 1980, this represents almost 16 percent of the total increase in average weeks worked over the time period.

To explore our last result in greater depth, we next pool our 45–50 year-old women two cohorts at a time, and examine the effect of the mobilization rate in explaining the variation in female labor supply in the later of the two cohorts. This approach allows us to examine the incremental effect of the war on the later of the cohorts under comparison. It also allows the state of birth effect to change with the pair of cohorts examined.

As shown in Table V, the pattern is similar to that obtained in Table IV. That is, there is a positive and significant effect of the war on 45–50 year-old women in 1950 (i.e., on the 1900–1905

TABLE V
IMPACT OF WORLD WAR II MOBILIZATION RATES ON LABOR SUPPLY
OF 45 TO 50 MARRIED WOMEN

Dependent variable is "Weeks Worked"				
	1940 & 1950	1950 & 1960	1960 & 1970	1970 & 1980
1940 mobilization rate × year	16.48** (7.87)	-11.66 (8.46)	-2.10 (6.28)	20.65*** (7.07)
1940 share male nonwhite × year	4.56 (4.91)	-9.00* (4.78)	-1.77 (4.54)	-1.81 (2.43)
1940 share male farmer × year	9.84*** (2.03)	-4.39* (2.42)	2.23 (1.86)	2.96 (2.55)
1940 male avg years educ × year	2.12*** (.436)	-.96** (.476)	-.46 (.576)	-.21 (.225)
Year	-22.35** (6.12)	20.93*** (7.62)	10.01 (6.33)	-4.37 (4.69)
N. obs.	15,955	25,127	45,015	45,037
Adjusted R^2	0.035	.029	.015	.017

Robust standard errors in parentheses account for clustering at the state-year level. Each column pools together 45 to 50 year-old women two cohorts at a time. Column 1 pools the 1900–1905 cohort in 1950 and the 1890–1895 cohort in 1940; column 2 pools the 1910–1915 cohort in 1960 and the 1900–1905 cohort in 1950; column 3 pools the 1920–1925 cohort in 1970 and the 1910–1915 cohort in 1960; column 4 pools the 1930–1935 cohort in 1980 and the 1920–1925 cohort in 1970. The dependent variable, weeks worked, is regressed on the mobilization rate variable (interacted with year dummies) assigned by the woman's state of birth. We also control for state fraction of male farmer, the state fraction of nonwhite male, and the state average education of males in 1940. All the 1940 variables (interacted with year dummies) are also assigned by the woman's state of birth. All specifications include state of birth dummies, year dummies, age dummies, and the latter interacted with a year dummy. Data are from Census IPUMS 1 percent sample for all years. For 1950 they refer to the sample line subsample. All specifications are weighted by census sampling weights. Our sample consists of 45 to 50 year-old white married women born in mainland United States excluding Nevada and DC, not living in institutional quarters, not living on farms or working in agricultural occupations. The sample is further restricted to women who were born in the same state they reside in. * Significance at 10 percent level. ** Significance at 5 percent level. *** Significance at 1 percent level.

cohort in 1950), there is no additional effect of the war in 1960 relative to 1950 (i.e., on the 1910–1915 cohort in 1960), there is no additional effect of the war in 1970 relative to 1960 (i.e., on the 1920–1925 cohort in 1970), and lastly there is a positive, significant, and quantitatively important additional effect of the war in 1980 relative to 1970 (i.e., on the 1930–1935 cohort in 1980).

The results presented in Table V demonstrate that not only does the variation in the mobilization rate help to explain the labor supply of 45–50 year-old women in 1980 relative to 1940 (as shown in Table IV), but it also helps explain the labor supply of women this age in 1980 relative to 1970. That is, at this point the indirect effect of the war is sufficiently large relative to the direct effect, that it significantly explains the labor supply of the 1930–1935 cohort relative to the 1920–1925 cohort. The value of 20 of

the mobilization coefficient in 1980 implies that a 10 percent higher mobilization rate is associated with a two-week increase in the average number of weeks worked by 45–50 year-old married women in 1980. Given that married women of this age were working on average 21.8 weeks in 1970, this represents around a 9 percent increase in the number of weeks worked.⁴⁷

We conclude from the evidence above that World War II directly affected the labor supply of older cohorts and indirectly affected a younger cohort. This younger cohort was too young to have changed its labor supply in direct response to the war. Nonetheless, it witnessed a permanent increase in its labor supply that varied across states with the mobilization rate of men. Our hypothesis is that this was a response to the increase in the number of men brought up by working mothers. The indirect effect of the mobilization rate on this particular cohort's labor supply is present at all points in the life cycle that we have examined, which also distinguishes it from the direct effect of the war which appears to fade as the earlier cohorts age.

Our analysis of the direct and indirect effects of the war also allows us to provide a rough estimate of the intergenerational effect that, according to our model, stems from having more sons brought up by working mothers. In particular, the ratio of the coefficient on the mobilization rate of the younger generation (the indirect effect) to the same coefficient for the mothers' generation (the direct effect), can be interpreted as the effect of an exogenous increase in the labor supply of married women on next generation's female labor supply. This ratio is 1.67, which implies that if married women of the age to have young children exogenously increase their labor supply by one week we expect an increase in the labor supply of the next generation of approximately 1.67 weeks.⁴⁸ It follows that the amplification mechanism deriving from the intergenerational channel is quantitatively large.⁴⁹

47. We find similar results when we run this regression exercise for the entire sample of white married women with and without dummies for state of residence and husband's state of birth.

48. We compute this number by dividing the coefficient obtained for the younger generation by 19.64 (the coefficient obtained for the older generation from Table III). The magnitude of the numerator depends on the age and specification, of course. Using the coefficient obtained when they are 45–50 years old in the third specification in Table IV, we get 1.67.

49. One should expect the size of this effect to decrease over time, as the economy approaches a steady state.

III.D. Alternative Interpretations

Our interpretation of the results obtained from our World War II analysis can be challenged by three alternative hypotheses. We next turn to a discussion of these and show that our explanation dominates these alternatives.

A first competing explanation is that the intergenerational effect we observe is brought about by working mothers affecting their daughters directly. A second hypothesis is that society was most transformed in those states with higher mobilization rates, making it easier for women to work in those states in the future. Although our dynamic empirical results alone cannot distinguish between our hypothesis and these alternatives, our cross-sectional evidence makes us feel more confident that our dynamic effect results, at least in large part, from the effect of working women on their sons. In particular, as we indicated previously, the effect of a woman's own mother working on the probability that she works full time when married appears to be negligible. Second, the large effect of a working mother on the probability that a man's wife works—from 24 to 32 percentage points—indicates that, in addition to any societal norms, the family plays an important role. Furthermore, as our result for women 45–50 years old demonstrates, the effect of World War II on the work behavior of women this age disappeared in the 1960s and 1970s (that is, for cohorts born in 1910–1915 and 1920–1925), only to resurface again in the 1980s for our 1930–1935 cohort. It is hard to think why changes in societal norms would give rise to this pattern.

A last possibility is that our dynamic results are really the consequences of the GI Bill. The GI Bill subsidized college education for World War II veterans. Male college enrollment jumped by more than 50 percent from the prewar (1939) level of 1.3 million to over 2 million men in 1946. Approximately one in eight veterans attended college. If the number of men attending college increased most in those states with the highest rate of mobilization, and if women “followed” men into college, then the positive correlation between the greater tendency of women from the 1930–1935 cohort to work and the mobilization rate of men across states could simply be a consequence of how this cohort increased its education differentially across states.

To examine the validity of this alternative hypothesis, we

perform the following two exercises. First, we examine the correlation between the increase in the average education of women in a state and that state's mobilization rate. Comparing the average education of white women born in a given state in 1920–1925 relative to that of those born a decade later in 1930–1935, we find that, averaging across states, education increased by .53 years, from 11.15 to 11.68 years.⁵⁰ The correlation between a state's change in average female education and its mobilization rate is negative and insignificant, independent of whether we assign education by a woman's current state of residence or whether we restrict our sample to women born in the same state in which they reside. Computing the partial correlation after controlling for the 1940 conditions in the state, in the same way as before, changes the sign of the correlation to positive, but likewise is statistically insignificant. Thus, the results of this first exercise make it doubtful that our findings are driven by the GI Bill.

To dispel any remaining doubts, we redid our World War II exercise controlling directly for a woman's level of education (an endogenous variable). Note that our theory also implies a positive relationship between mobilization rates and female education: a woman is likely to find additional education more attractive if she is planning to work in the market.⁵¹ Thus, even if once we controlled for education, the effect of mobilization rates on working became insignificant, this in itself would not be evidence against our theory. On the other hand, finding a significant effect of the mobilization rate variable on female labor supply even after controlling for education is evidence in favor of our theory. It shows that the effect of the mobilization rate does not go solely through education, which it would in the case of the GI Bill.

The last column in Table IV repeats the regression for the specification in column (iii), but also includes a set of education dummies as well as these dummies interacted with a year dummy. Education is measured as the highest grade of school

50. We calculate women's education in the decade the cohort reaches 35–40 years, i.e., 1960 and 1970, respectively, for the earlier and later cohort. Education is measured as the highest grade of school attended or completed by the respondent. This variable is topcoded at six years of college education (so all individuals with more than six years of college are assigned eighteen years of education).

51. Of course, market skills and education are not synonymous. In fact, when we control for female education in our cross-sectional regression, we still find that men with working mothers are more likely to work. Hence the effect does not run solely through education.

attended or completed by the respondent and falls into one of nine possible categories.⁵² As can be seen, the results are very similar to the ones in the previous column, with exactly the same pattern of positive and significant results, and quantitatively similar magnitudes. This allows us to conclude that increased education is not what is driving our results.

III.E. Fertility Ratio

An interesting implication of our theory is that, *ceteris paribus*, states in which the ratio of children brought up by working women relative to nonworking women is greater, should have greater female labor supply in the next generation. This follows from the fact that, everything else equal, the larger the average fertility of working relative to nonworking women (hereafter denoted the “fertility ratio” for short), the larger the proportion of men in the following generation whose mothers worked. If our theory is correct, this should make investing in market skills more attractive for women in the next generation, thereby increasing female labor supply.

In this section we examine the relationship between the fertility ratio across states and female labor supply twenty years later. We regress various measures of female labor supply on a set of individual-level characteristics (age and marital status) and on two state level variables that are assigned to women by their state of birth. For this exercise we restrict attention to white women whose state of birth coincides with her state of residence and pool data from the 1960, 1970, and 1980 Census. We use the following specification:

$$L_{ist} = X'_{ist}\beta_1 + \alpha_1 L_{st-20} + \alpha_2 \left(\frac{f_w}{f_n} \right)_{st-20} + d_s + \gamma_t + \varepsilon_{ist}.$$

In this regression L_{ist} measures the labor supply of a 25–30 year-old woman born in state s at time t . X_{ist} represents a set of individual characteristics: age dummies and marital status dummies and both variables interacted as well with time dummy. All the regressions also include state of birth dummy d_s and a time dummy γ_t .

52. The nine categories are none or preschool, grade 1 to 4, grade 5 to 8, grade 9, grade 10, grade 11, grade 12, 1 to 3 years of college, 4 plus years of college.

There are two state-level variables. The first, L_{st-20} , is the twenty-year-lagged average labor supply of women 30 to 35 years old, assigned by the individual's state of birth. This variable is introduced in order to control for the "initial" level of female labor supply in each state. To examine our thesis, we used three alternative definitions of labor supply. The first, labor force participation (LFP), is an indicator variable that takes the value of one if a woman was in the labor force in the week before the interview (Census definition) and equals zero otherwise. The second, Positive Hours, is an indicator variable that equals one if a woman worked a positive number of hours over the past week, and equals zero otherwise. Lastly, Weeks Worked is the same variable we used in our World War II analysis: it indicates how many weeks a woman worked in the previous year.

The second state-level variable, $(f_w/f_n)_{st-20}$, is our variable of interest. It is the twenty-year-lagged ratio of the average fertility of working women relative to that of nonworking women in the individual's state of birth. This variable is calculated as the ratio of the average number of their own children living in the households of 30–35 year-old working women (f_w) relative to the same average for 30–35 year old nonworking women (f_n). The definition of a "working woman" used to construct the fertility ratio varies to concord with the definition we are using for the dependent variable (and for L_{st-20} as well).⁵³ We expect that, conditional on the same level of female labor supply, states characterized by a higher relative fertility ratio of working to nonworking women at a point in time should also be characterized by a higher labor supply of women twenty years later.

It is interesting to note that, independently of the definition of working women adopted, the average fertility ratio has been increasing over time for all definitions of working women. It went from an average across states of 0.34 in 1940 to 0.62 in 1960. Furthermore, at any point in time the variance in fertility ratios across states is quite large. In 1940 the fertility ratio ranged from a minimum of .18 in Montana to a maximum of .63 in South

53. For the definition "weeks worked" we used whether a woman had worked a positive number of weeks in the previous year, however, as the first is a continuous variable. Similar results were obtained when we used alternative definitions to construct the fertility ratio.

TABLE VI
FERTILITY REGRESSIONS

	Dependent variable is		
	LFP	Positive hours	Weeks worked
Avg Fertility Ratio t-20	.126*** (.047)	.104** (.046)	4.53** (2.33)
N. obs.	129341	129341	129341
Adjusted R^2	0.18	0.16	0.18

Robust standard errors in parentheses account for clustering at the state-year level. Each column is for a separate pooled regression for the years 1960 to 1980 of different measures of labor force participation for women 25–30 at time t , on the average labor force participation of women 30–35 twenty years before assigned by state of birth and on the ratio of average fertility of working women age 30–35 over average fertility of nonworking women age 30 to 35 twenty years before, also assigned by state of birth. The first column uses the Census definition of LFP; the second column defines work as an indicator variable that equals 1 if a woman worked positive hours during the week before the interview, and the last column uses the number of weeks worked in previous year as dependent variable. Average fertility is defined as average number of own children in the household, and the definition of working versus nonworking woman changes across columns according to the definition adopted for the dependent variable in each specification. In the last column a woman is defined as working if she worked a positive number of weeks in the previous year. Our specification also includes a constant, state-fixed effects, year main effects, age and marital status dummies, and their interaction with 1960, 1970, and 1980 dummies. Data are from Census IPUMS 1 percent sample for all years. For 1950 they refer to the sample line subsample. Our sample consists of white women residing in mainland United States, not living in institutional quarters, not living on farms or working in agricultural occupations.

* Significance at 10 percent level. ** Significance at 5 percent level. *** Significance at 1 percent level.

Carolina; by 1960 it ranged from a minimum of .28 in the District of Columbia and Delaware to a maximum of .78 for Mississippi. The mean across states and time periods is .49 with a standard deviation of .13.⁵⁴

Table VI reports the results of our regression analysis. For all definitions of working women, we find a positive and significant relationship between women's working behavior and the average fertility ratio of working relative to nonworking women twenty years earlier. The magnitude of the effect of fertility on future female labor supply seems to be very similar across all definitions. In particular, an increase by one standard deviation in the average fertility ratio is associated with, twenty years later, an increase of 1.7 percentage points in LFP (i.e., an increase of about 3.5 percent over its sample mean of 44 percent), an increase of .58 weeks per year in weeks worked (i.e., an increase of 3.3 percent over its sample mean of 17.6 weeks), or with an increase in the

54. The numbers given here are for the LFP definition of a working woman. Similar means and variances are obtained using the alternative definitions.

proportion of women with positive hours of 1.4 percentage points (an increase of around 3.2 percent over its sample mean).⁵⁵

An important shortcoming of our analysis, of course, is that we are unable to identify an exogenous source of variation in the fertility ratio.⁵⁶ Nonetheless, the positive correlation between the fertility ratio and female labor force participation twenty years later constitutes suggestive evidence that favors our hypothesis.

IV. CONCLUSION

Over the last century, society has been deeply transformed: not only is woman's economic role radically different, but a new family model has developed, and individuals' expectations and preferences toward marriage and gender roles have evolved in important ways.⁵⁷ Standard explanations for the changed economic role of women over time have focused primarily on technological factors: the introduction of time-saving consumer durables that reduced the time required to carry out traditional tasks in the household, the advent of the pill that enabled women to control fertility, and the shift toward a service and skill-intensive economy that increased the proportion of jobs suitable for women.

Our paper develops the idea that the greater presence over time of a new type of man—one brought up in a family with a working mother—contributed significantly to increasing female labor supply over time. We provide cross-sectional and dynamic evidence for the general thesis that family attitudes and their intergenerational transmission played a quantitatively significant role in transforming women's role in the economy. In particular, a working mother appears to affect her son's preferences (or abilities) in a way that has important consequences for his wife's working behavior, making it more likely that she

55. Our results are similar if we do not restrict the sample to women who reside in the same state as their state of birth.

56. It may be argued that the same factors that cause the fertility ratio to be higher in one state relative to another, may also make it more attractive for young women to work more in that state twenty years later. The simplest version of this critique, however, is taken care of by controlling for the state's lagged female labor supply alongside its lagged fertility rate (and by including a state fixed effect). Hence, if, for example, one state has better child-care services than another, making it easier for women both to work and to have children, this should be captured by controlling for female labor supply.

57. This evolving role of women is still a topic of hot debate, especially the extent to which professional women are able to successfully combine children and a career (see, e.g., Hewlett [2002]).

works. We construct a model that allows us to study the dynamic consequences of the endogenous increase in the proportion of these men. As we show, our model generates cross-sectional and dynamic implications that are consistent with the data.

Using several data sets, we show that the probability that a man's wife works is positively and significantly correlated with whether his mother worked, even after controlling for many other background characteristics of husband and wife such as religion, geography, family wealth, and whether the wife's mother worked. We find that having a working mother significantly increases the probability that a man's wife works; the magnitude of the effect ranges from 24 to 32 percentage points, depending on the definition of a working mother and the data set used.

According to our theory, an event that increases the proportion of men brought up by working mothers has dynamic repercussions on women's labor supply since it makes investment in market skills and work more attractive to women in the next generation. We use variation in the World War II mobilization rates of men across the United States to provide exogenous variation in the magnitude of the shock that the war provided to female labor supply. We show that the mobilization rate had a positive impact on the labor supply of women most likely to have young children during this period. This impact of World War II finds an echo many decades later in the labor supply of women who were too young to have been directly affected by the war, but who would have been affected by the implied change in the composition of the future marital pool—the cohort born in 1930–1935. We show that whereas the direct effect of the war on older cohorts fades over time, the indirect effect of the war persists in the work behavior of the 1930–1935 cohort during its life cycle. We also provide a rough estimate of the magnitude of the intergenerational channel: we find that an increase of one week in the average female labor supply leads to an increase of 1.67 weeks worked in next generation's female labor supply.

Our model also implies that the greater the average fertility of working relative to nonworking women, *ceteris paribus*, the greater should be female labor supply next generation. We examine this relationship across states for several decades, and show that a positive correlation exists in the data, even after controlling for the state's initial labor supply and a state fixed effect.

We consider our paper to be a contribution to a small but growing literature that is interested in examining how attitudes (or preferences), social norms, or culture influence the evolution of the economy. We are especially interested in attempting to assess the quantitative significance of what are often considered to be rather “fuzzy” variables, whose existence is often implicitly recognized but rarely quantified. These variables, however, may play a significant role in many economic phenomena, from female education and labor dynamics to fertility, consumption, and investment decisions, and thus are too important to be neglected. In particular, we have shown that there is a quantitatively important link between the proportion of men with working mothers and women’s propensity to work. We think that studying the evolution of the family and its interaction with the economy may be fertile ground for future research in this area.

APPENDIX 1: DESCRIPTIVE STATISTICS: GSS AND FLFPMI DATA

Variable	GSS		FLFPMI	
	Mean	S.D.	Mean	S.D.
WIFEWOR	.53	.50	.45	.50
MAWORKH	.50	.50	.12	.33
MAWORKW			.09	.29
HUSB AGE	41.0	6.4	35.9	9.0
HUSB EDUC	14.4	3.0	14.5	2.7
HUSB INCOME	33.4	21.2	23.0	13.5
WIFE AGE	38.0	5.7	33.8	8.9
WIFE EDUC	13.6	2.7	13.8	2.2
CHILDREN	2.2	1.3	1.9	1.5
MAEDUCH	11.3	3.1	11.5	3.0
PAEDUCH	11.1	3.8	11.4	3.7
BABIES	.37	.69		
MAEDUCW			11.7	2.8
PAEDUCW			11.6	3.6
N. obs.	189		969	

Source: GSS 1988 and 1994 and FLFPMI 1980. The GSS sample consists of all white married men whose wives are from 30 to 50 years old. For GSS sample, WIFEWOR = 1 if wife employed full time, or temporarily away from job because of illness, vacation, or strike during the week preceding the interview. For FLFPMI sample, WIFEWOR = 1 if, at the time of the interview, the wife was working for pay in a full-time job. For GSS sample, MAWORK = 1 if husband’s mother ever worked for pay for as long as one year after he was born and before he was fourteen. For FLFPMI sample, MAWORK = 1 if the husband’s mother worked all the time while her son was growing up. The GSS data set also includes nine regional dummies, six residential dummies, four religion dummies, and three dummies for husband’s self-assessment of family income at age sixteen. The FLFPMI data set also includes nine regional dummies, three residential dummies, four religion dummies, and two continuous variables capturing the socioeconomic status of the fathers of husband and wife.

APPENDIX 2: SUMMARY CHARACTERISTICS OF MARRIED WOMEN, 1940–1980

	1940	1950	1960	1970	1980
<i>25 to 30 years old</i>					
Weeks Worked	8.83 (17.98)	10.25 (18.29)	11.78 (18.77)		
Age	27.02 (1.41)	27.02 (1.41)	27.04 (1.42)		
Husband's age	31.10 (5.22)	30.55 (4.75)	30.40 (4.43)		
Number of observations	27,145	11,702	34,862		
<i>35 to 40 years old</i>					
Weeks Worked	7.44 (16.95)	10.79 (19.09)	14.01 (20.49)	18.02 (21.81)	
Age	36.94 (1.43)	36.92 (1.41)	36.98 (1.41)	37.01 (1.42)	
Husband's age	40.81 (5.64)	40.60 (5.33)	40.30 (5.12)	40.19 (5.01)	
Number of observations	22,979	10,153	41,595	35,137	
<i>45 to 50 years old</i>					
Weeks Worked	5.52 (15.00)	11.81 (19.97)	17.91 (22.18)	21.83 (23.05)	26.26 (23.59)
Age	46.91 (1.40)	46.95 (1.42)	46.90 (1.41)	46.96 (1.41)	47.03 (1.43)
Husband's age	50.32 (5.64)	50.57 (5.76)	50.25 (5.69)	49.92 (5.41)	50.11 (4.88)
Number of observations	16,717	6,922	32,007	38,082	33,999

Standard errors are in parentheses. Data are from the Census IPUMS 1 percent sample for 1940 to 1980. Data for 1950 refer to the sample line subsample. The sample consists of white married women belonging to three age groups: 25–30, 35–40, and 45–50. We exclude women living on farms or employed in farming, women living in institutional group quarters, and women who were born in Alaska, Hawaii, Nevada, D.C., or abroad.

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REFERENCES

- Acemoglu, Daron, David Autor, and David Lyle, "Women, War, and Wages: The Effect of Female Labor Supply on the Wage Structure at Midcentury," *Journal of Political Economy*, CXII (2004), 497–551.
- Akerlof, George A., and Rachel E. Kranton, "Economics and Identity," *Quarterly Journal of Economics*, CXV (2000), 715–753.
- Bisin, Alberto, and Thierry Verdier, "Beyond the Melting Pot: Cultural Transmission, Marriage, and the Evolution of Ethnic and Religious Traits," *Quarterly Journal of Economics*, CXV (2000), 955–988.
- Blau, Francine D., Marianne Ferber, and Anne Winkler, *The Economics of Women, Men, and Work*, 4th ed. (Upper Saddle River, NJ: Prentice Hall, 2002).
- Booth, Alan, et al., "Female Labor Force Participation and Marital Instability,"

- Inter-university Consortium for Political and Social Research, Study No. 9199. <http://www.icpsr.umich.edu:8080/ICPSR-STUDY/09199.xml>, 1980.
- Cunningham, Mick, "Parental Influences on the Gendered Division of Household Work," *American Sociological Review*, LXVI (2001), 184–203.
- Daly, Martin and Margo Wilson, "Is Parent-Offspring Conflict Sex-Linked? Freudian and Darwinian models," *Journal of Personality*, LVIII (1990), 163–189.
- Davis, James A., and Tom W. Smith, *General Social Surveys, 1972–1998: Cumulative Codebook* (Chicago: NORC, 1999).
- Del Boca, Daniela, Marilena Locatelli, and Silvia Pasqua, "Employment Decisions of Husbands and Wives," *Labour*, XIV (2000), 35–52.
- Duncan, Otis D., "A Socioeconomic Index for All Occupations," in A. Reiss et al., *Occupations and Social Status* (New York: Free Press, 1961).
- Epstein, Elizabeth and Ruth Guttman, "Mate Selection in Man: Evidence, Theory, and Outcome," *Social Biology*, XXXI (1984), 243–278.
- Ersine, Hazel, "The Polls: Women's Role," *Public Opinion Quarterly*, XXXV (1971), 275–290.
- Fernández, Raquel, Alessandra Fogli, and Claudia Olivetti, "Marrying Your Mom: Preference Transmission and Women's Labor and Education Choices," NBER Working Paper No. 9234, 2002.
- Fernández, Raquel, Alessandra Fogli, and Claudia Olivetti, "Preference Formation and the Rise of Women's Labor Force Participation: Evidence from WWII," NBER Working Paper No. 10589, 2004.
- Fernández, Raquel, Nezih Guner, and John Knowles, "Love and Money: A Theoretical and Empirical Analysis of Household Sorting and Inequality," NBER Working Paper No. 8580, 2001.
- Fernández, Raquel, and Richard Rogerson, "Sorting and Long-Run Inequality," *Quarterly Journal of Economics*, CXVI (2001), 1305–1341.
- Fogli, Alessandra, "Labor Market Rigidities and Family Ties," Ph.D. thesis, University of Pennsylvania, 2000.
- Freud, Sigmund, *Some Psychological Consequences of the Anatomical Distinction Between the Sexes*, in J. Strachey, ed. and Trans. Standard Edition (Vol. 8, pp. 133–142) (London: Hogart Press, 1927), pp. 133–142.
- Galor, Oded, and Omer Moav, "Natural Selection and the Origin of Economic Growth," *Quarterly Journal of Economics*, CXVII (2002), 1133–1192.
- Galor, Oded, and David N. Weil, "The Gender Gap, Fertility and Growth," *American Economic Review*, LXXXVI (1996), 374–387.
- Goldin, Claudia, *Understanding the Gender Gap: An Economic History of American Women* (New York: Oxford University Press, 1990).
- , "The Role of WWII in the Rise of Women's Employment," *American Economic Review*, LXXXI (1991), 741–756.
- , "Career and Family: College Women Look at the Past," in *Gender and Family in the Workplace*, F. Blau and R. Ehrenberg, eds. (New York: Russell Sage Press (1997), pp. 20–58.
- Goldin, Claudia and Lawrence F. Katz, "The Power of the Pill: Oral Contraceptives and Women's Career and Marriage Decisions," *Journal of Political Economy*, C (2002), 730–770.
- Greenwood, Jeremy, Ananth Seshadri, and Mehmet Yorukoglu, "Engines of Liberation," *Review of Economic Studies*, (2004), forthcoming.
- Hewlett, Sylvia Ann, *Creating a Life: Professional Women and the Quest for Children* (New York: Hyperion, 2002).
- Jones, Larry, Rodolfo E. Manuelli, and Ellen R. McGrattan, "Why Are Married Women Working So Much?" Research Department Staff Report 317, Federal Reserve Bank of Minneapolis, 2003.
- Killingsworth, Mark R., and James J. Heckman, "Female Labor Supply: A Survey," Chapter 2 in *Handbook of Labor Economics*, Volume 1, O. Ashenfelter and R. Layard, eds. (New York: North-Holland, 1986).
- Kremer, Michael, "How Much Does Sorting Increase Inequality?" *Quarterly Journal of Economics*, CXII (1997), 116–139.
- Lundberg, Shelly J., and Robert Pollak, "Efficiency in Marriage," NBER Working Paper 8642, 2001.
- National Center for Health Statistics, "Live Birth of Mother and Race: United States, 1933–98," <http://www.cdc.gov/nchs/data/natality/mage33tr.pdf>, 2002.

- Neumark, David, and Andrew Postlewaite, "Relative Income Concerns and the Rise in Married Women's Employment," *Journal of Public Economics*, LXX (1998), 157–183.
- Olivetti, Claudia, "Changes in Women's Hours of Market Work: The Effect of Changing Returns to Experience," Ph.D. thesis, University of Pennsylvania, 2001.
- Pencavel, John H., "The Market Work Behavior and Wages of Women, 1975–94," *Journal of Human Resources*, XXXVIII (1998), 771–804.
- Smith, James P., and Michael P. Ward, "Time-Series Growth in the Female Labor Force," *Journal of Labor Economics*, III (1985), 59–90.
- Thornton, Arland, Duane F. Alwin, and Donald Camburn, "Causes and Consequences of Sex-Role Attitudes and Attitude Change," *American Sociological Review*, XLVIII (1983), 211–227.